Incomplete, Correlated and Bayesian Data Analysis

Course manual: Lectures 1–3 and complementary notes on linear mixed models

Universidad de Alcalá

11 November 2026

Abstract

This manual reorganises the slides of a three-day course into a single, ordered text. **Part I** (Lecture 1) treats data analysis with incomplete observations: missing-data mechanisms, observed-data likelihood and the EM algorithm. **Part II** (Lecture 2) introduces longitudinal data and linear, generalised and nonlinear mixed models, with an extended R application. **Part III** collects the complementary slides on correlated data and linear mixed models (marginal models, covariance structures, likelihood inference and testing). **Part IV** (Lecture 3) covers Bayesian inference, MCMC (Metropolis–Hastings, Gibbs, HMC), JAGS and Stan, with the Berkeley admissions example. Text, formulas, tables, R code and the numerical outputs printed on the slides are reproduced, together with the figures: 17 plots and diagrams are cropped directly from the original slides (keeping their original colours), while the covariance-structure tables and the `nlme` example that appear as images in the slides are retypeset. Passages marked "Note (added)" are not on the slides.

Contents

Part I. Data Analysis with Incomplete Observations

Background and learning objectives

Ideal data and real data

Two naïve remedies

Likelihood inference with complete data

Incomplete covariate data: what changes

Missing-data mechanisms

Likelihood inference with incomplete covariate data

The EM algorithm

Worked application: a salary survey

R application: simulated salary survey

Recap, take-home messages and references

Part II. Cross-sectional and Longitudinal Data Analysis with Latent Variable Models

Background and learning objectives

Running example: student grades and study hours

The linear mixed-effects model

Likelihood inference for the linear mixed model

Application: a panel of 800 students

Model comparison

Model checking

Student-specific estimates, shrinkage and prediction

Generalised and nonlinear mixed models
Other correlation structures, workflow and take-aways

Part III. Complementary Notes: Correlated Data and Linear Mixed Models

Preface and course context
Introduction: correlated data
Why specific analytic tools? Two examples
Linear mixed models and marginal models
Correlated data and longitudinal studies
The efficiency of longitudinal designs
Linear mixed models
Covariance models for clustered data
Likelihood inference for linear mixed models

Part IV. Bayesian Statistics: Models, Inference and Computation

Frequentist versus Bayesian inference
Application: estimating an admission probability
Bayes' rule, conjugacy and the Beta–Binomial model
A model for admission by sex and department
Markov chain Monte Carlo
JAGS: Just Another Gibbs Sampler
Gibbs sampling
Hamiltonian Monte Carlo, NUTS and Stan
Choosing a computational approach and practical workflow
Further topics: large datasets and variational inference

Part I

Data Analysis with Incomplete Observations

Lecture 1 of 3 · F. Ungolo, UNSW Sydney

Background and learning objectives

Statistical methods for the analysis of data (collection, probabilistic modelling, estimation and prediction) have been developed primarily for working with *rectangular datasets*. Suppose we want to analyse the voting pattern of Spanish people; for each person we may observe age, sex, religion, education, household income band (a proxy for socio-economic position), policy preference or vote choice, ethnicity, and so on. In practice, however:

- data-collection methods may fail to fully collect individual information;
- individuals may not be able (or willing) to answer a question, for instance in opinion polls.

Learning objectives

- Data collection and availability: ideal versus real;
- statistical mechanisms generating missing observations;
- likelihood inference with missing observations;
- the EM algorithm;
- application in R.

Ideal data and real data

In the ideal situation every respondent has a value for every variable (Table 1, with n respondents). In reality some cells are empty (Table 2, a poll of ten respondents in which the income band is unrecorded for units 6–10).

Table 1. Analysis of voting preferences: an ideal situation (rectangular, complete data).

Unit	Age	Gender	Region	Education	Income band	Vote choice
1	25	F	Madrid	University	High	Party A
2	42	M	Andalucía	Secondary	Medium	Party B
3	43	F	Cataluña	University	Low	Party A
⋮	⋮	⋮	⋮	⋮	⋮	⋮
n	55	F	Galicia	Secondary	Medium	Party C

Table 2. Analysis of voting preferences: the reality (ten respondents; "–" denotes a missing value).

Unit	Age	Gender	Region	Education	Income band	Vote choice
1	25	F	Madrid	University	High	Party A
2	42	M	Andalucía	Secondary	Medium	Party B
3	43	F	Cataluña	University	Low	Party A
4	35	F	Madrid	University	Medium	Party B
5	28	M	Cataluña	Secondary	Low	Party C
6	39	F	Madrid	University	–	Party A
7	35	F	Madrid	University	–	Party B
8	28	M	Madrid	Secondary	–	Party C
9	46	M	Andalucía	University	–	Party B
10	55	F	Galicia	Secondary	–	Party C

When collecting data we may encounter the raw values of Table 3. Part of the work of the analyst is to understand which of them really denote a missing value.

Table 3. Raw values that may hide a missing observation.

Raw value	Possible reason
blank in income band	the interviewer did not record the variable, the field was lost while combining files, or the respondent skipped a sensitive income question
-99	a placeholder for "unknown"
N/A	not applicable for this person
0	a true zero, or a badly encoded blank
refused	a response behaviour

Two naïve remedies

Reduced dataset 1: complete cases

The first solution is to retain only respondents with a recorded value for every variable (*complete-case analysis*). In the poll above only units 1–5 survive.

Consequences. A smaller dataset is retained, hence the statistical uncertainty is higher. If there is a systematic pattern in the missing observations, the resulting analysis can be biased. For example, if people with low educational attainment tend to skip the income question, then the analysis can be biased if the remaining high-education individuals tend to vote for a certain party.

Reduced dataset 2: common variables

The second solution is to keep all respondents but remove the variables with missing entries (here, income band is dropped and all ten respondents remain with age, gender, region, education and vote choice).

Consequence. Statistically significant information may be lost.

Likelihood inference with complete data

Regression models

Let Y denote the dependent variable and X the vector of covariates; the interest lies in the parametric density $f_\theta(y \mid x)$. For example, the linear regression model

$$y_i = \alpha + \beta^\top x_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2),$$

with unknown parameter vector $\theta = (\alpha, \beta, \sigma)$. For a linear model with normally distributed error terms, the density of y conditional on x is

$$f_\theta(y \mid x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{ -\frac{1}{2}\Big(\frac{y - \alpha - \beta^\top x}{\sigma}\Big)^2 \Big\}.$$

Maximum likelihood with fully observed data

If data are fully observed, θ can be estimated by maximum likelihood,

$$\hat\theta = \arg\max_\theta L(\theta \mid y, x), \qquad L(\theta \mid y, x) = \prod_i f_\theta(y_i \mid x_i).$$

For numerical and computational convenience we (equivalently) maximise the logarithm of the likelihood, $\log L(\theta \mid y, x) = \sum_i \log f_\theta(y_i \mid x_i)$. "Equivalent" means here that the maximisation of the two functions yields the same value of the argument θ.

If we are also interested in the distribution of the covariates, $f_\xi(x)$, then

$$(\hat\theta, \hat\xi) = \arg\max_{(\theta,\xi)} \log L(\theta, \xi \mid y, x), \tag{1}$$

where

$$L(\theta, \xi \mid y, x) = \prod_{i=1}^n f_\theta(y_i \mid x_i) f_\xi(x_i) = \underbrace{\prod_{i=1}^n f_\theta(y_i \mid x_i)}_{L_Y(\theta \mid y, x)} \underbrace{\prod_{i=1}^n f_\xi(x_i)}_{L_X(\xi \mid x)}.$$

Result 1 (complete-data estimation). *With distinct parameters, $\hat\theta$ and $\hat\xi$ can be estimated by maximising the two factors separately.*

Incomplete covariate data: what changes

For the ith unit in the sample, partition the vector of covariates as $x_i = (x_{i,\text{obs}}, x_{i,\text{mis}})$. In the longer displays below we abbreviate $x_i^o = x_{i,\text{obs}}$, $x_i^m = x_{i,\text{mis}}$ and $\mathcal{X}_m = \mathcal{X}_{\text{mis}}$. Three consequences follow:

1. we need to work with the marginal distribution of the (observable) data;
2. we need to account for the probability distribution of the covariates $f_\xi(x_i)$ in order to estimate θ (at least those covariates subject to missingness across the sample, i.e. missing for at least one unit);
3. we need to account for the missing-data mechanism.

The missingness indicator

For unit i, let $X_i = (X_{1,i}, \ldots, X_{p,i})^\top$ be the set of covariates and define

$$M_{j,i} = \begin{cases} 1, & \text{if } X_{j,i} \text{ is observed,} \\ 0, & \text{if } X_{j,i} \text{ is missing,} \end{cases} \qquad j = 1, \ldots, p,$$

together with a probabilistic model for $M_i = (M_{1,i}, \ldots, M_{p,i})$:

$$f_\psi(m_i \mid y_i, x_i). \tag{2}$$

The key issue is: *why are data missing?*

Missing-data mechanisms

Missing-data mechanisms are of three types, according to the taxonomy of Rubin (1976).

Definition 2 (MCAR: missing completely at random). The probability distribution of the missing-data mechanism does not depend on the value of the missing covariate, nor on the value of other covariates, nor on the outcome y_i:

$$f_\psi(m_i \mid x_i, y_i) = f_\psi(m_i).$$

Definition 3 (MAR: missing at random). Missingness may depend on information already observed, but not on the missing value after conditioning on that information:

$$f_\psi(m_i \mid x_i, y_i) = f_\psi(m_i \mid x_{i,\text{obs}}, y_i).$$

Definition 4 (MNAR: missing not at random). The probability distribution of M_i depends on the value of $x_{i,\text{mis}}$ (and can also depend on $x_{i,\text{obs}}$ and y_i). The probability of not being able to observe the value of a covariate depends on the value of the covariate itself.

Table 4. The three mechanisms with the examples used in the lecture and their implication for likelihood inference.

Mechanism	Examples	Implication for the likelihood
MCAR	*Spanish voting survey:* a randomly selected batch of interviewers does not fill in the household-income field, as it is optional; the probability of filling it in does not depend on age, region, education or vote. *Weekly fitness training hours:* the outcome depends on working hours, sleeping hours and accumulated fatigue; fatigue is missing for individuals who forget to wear a smartwatch.	Ignorable for the likelihood. Complete-case maximum likelihood is valid when the model is correct, but it discards information.
MAR	*Spanish voting survey:* respondents skip the income question more often in some survey modes or regions; after conditioning on observed mode, region, age, education and vote choice, omission does not depend on the unobserved income band. *Salary survey:* respondents skip the education question more often online than by telephone; after conditioning on observed mode and experience, omission does not depend on the unreported education.	With distinct parameters, the missingness model can be ignored. Use the observed-data likelihood, which integrates over the missing values.
MNAR	*Spanish voting survey:* even after accounting for observed age, education, region and vote choice, respondents with very high household income are more likely to skip the income question. *Household budget survey:* even after accounting for household size and region, high-income households are more likely to refuse the income-band question.	The missingness model remains in the likelihood. Estimate it jointly with the outcome model, or carry out a sensitivity analysis.

Discussion exercises

Three discussion questions are posed on the slides. They all refer to a Spanish voting poll, collected online and by telephone, that records voting preference and age, sex, level of education, income, home ownership and type of job.

1. *Discussion 1.* The income information is more likely to be missing in online interviews.
2. *Discussion 2.* Interviewees with a PhD tend not to disclose the information on their income.
3. *Discussion 3.* Interviewees with a high income tend not to disclose this information.

Note (added). Suggested classification, which the slides leave to the discussion: (1) MAR, because the interview mode is observed; (2) MAR, because the level of education is observed and missingness depends on it, not on the unseen income; (3) MNAR, because missingness depends on the value of income itself.

Likelihood inference with incomplete covariate data

The joint model and the observed-data likelihood

For the ith unit, suppose we are interested in a predictive regression model $f_\theta(Y_i \mid x_i)$. When covariates can be missing, three variables need to be considered, Y_i, X_i and M_i, and the joint probability model has parameter vector $\gamma = (\theta, \xi, \psi)$:

$$f_\gamma(Y_i, X_i, M_i) = f_\theta(Y_i \mid X_i, M_i)\, f_\psi(M_i \mid X_i)\, f_\xi(X_i) = f_\gamma(Y_i \mid X_i)\, f_\psi(M_i \mid X_i)\, f_\xi(X_i).$$

The marginal density of the observable data $(y_i, x_{i,\text{obs}}, m_i)$, which is the likelihood contribution of unit i, is then

$$\begin{aligned} L_i(\theta, \psi, \xi) &= \int_{\mathcal{X}_m} f_\theta(y_i \mid x_i) f_\psi(m_i \mid x_i) f_\xi(x_i)\, dx_i^m \\ &= \int_{\mathcal{X}_m} f_\theta(y_i \mid x_i^o, x_i^m) f_\psi(m_i \mid x_i^o, x_i^m) f_\xi(x_i^o, x_i^m)\, dx_i^m. \end{aligned}$$

When data are missing, the maximum likelihood estimator still has the same asymptotic properties as when data are completely observed (Takai & Kano, 2013):

$$(\hat\theta, \hat\xi, \hat\psi) = \arg\max_{(\theta,\xi,\psi)} \log L(\theta, \xi, \psi \mid y, x_{\text{obs}}, m) = \arg\max_{(\theta,\xi,\psi)} \sum_i \log L_i(\theta, \xi, \psi).$$

Two key issues.

- The individual contributions

$$\log L_i(\theta, \xi, \psi) = \log \int_{\mathcal{X}_{\text{mis}}} f_\theta(y_i \mid x_i) f_\psi(m_i \mid x_i) f_\xi(x_i)\, dx_{i,\text{mis}} \tag{3}$$

  imply that the likelihood of θ is *not separable* from that of ξ and ψ;
- the resulting log-likelihood given the observable data is not easy to optimise numerically, especially if the missing variable is continuous.

First simplification: an ignorable mechanism

If data are MCAR, then $f_\psi(m_i \mid x_i) = f_\psi(m_i)$ does not depend on the integration variable and

$$\log L_i(\theta, \xi, \psi) = \log \left[f_\psi(m_i) \int_{\mathcal{X}_{\text{mis}}} f_\theta(y_i \mid x_i) f_\xi(x_i)\, dx_{i,\text{mis}} \right].$$

The slides leave two questions open: *what if data are MAR? And when data are MNAR?*

Note (added). Under MAR, $f_\psi(m_i \mid x_i, y_i) = f_\psi(m_i \mid x_{i,\text{obs}}, y_i)$ is also constant in $x_{i,\text{mis}}$, so it factors out of the integral in the same way (the next slide writes it as $f_\psi(m_i \mid x_{i,\text{obs}})$) and, with distinct parameters, (θ, ξ) can be estimated separately from ψ. Under MNAR the factor $f_\psi(m_i \mid x_i)$ depends on $x_{i,\text{mis}}$ and must stay inside the integral.

Second simplification: discrete missing covariates

If the missing covariate(s) are discrete the integral becomes a sum,

$$\log L_i(\theta, \xi, \psi) = \log \Big\{ \sum_{x_{i,\text{mis}}} f_\theta(y_i \mid x_i) f_\psi(m_i \mid x_i) f_\xi(x_i) \Big\},$$

and, in case of an ignorable mechanism (MCAR and MAR),

$$\log L_i(\theta, \xi, \psi) = \log \Big\{ f_\psi(m_i \mid x_{i,\text{obs}}) \sum_{x_{i,\text{mis}}} f_\theta(y_i \mid x_i) f_\xi(x_i) \Big\},$$

which means that (θ, ξ) can be estimated separately from ψ.

The EM algorithm

Motivation

One possibility is to maximise the log-likelihood of the observable data directly. Is it the only available option? Dempster, Laird and Rubin (1977) published "Maximum likelihood from incomplete data via the EM algorithm" in the *Journal of the Royal Statistical Society, Series B*, where they formalised the Expectation–Maximisation (EM) algorithm. It was a massive breakthrough in computational statistics (nearly 80,000 citations according to Google Scholar) and is nowadays a core tenet of modern machine learning.

The idea

Instead of maximising $\sum_i \log L_i(\theta, \xi, \psi)$, Dempster et al. (1977) proved that we can find (θ, ξ, ψ) (or just (θ, ξ) if the missing-data mechanism is ignorable) by maximising the expected complete-data log-likelihood

$$\begin{aligned} \mathbb{E}\big[\log L(\theta, \xi, \psi \mid y, x, m)\big] = \sum_i \Big\{ & \mathbb{E}[\log f_\theta(y_i \mid x_i^o, x_i^m)] + \mathbb{E}[\log f_\xi(x_i^o, x_i^m)] \\ & + \mathbb{E}[\log f_\psi(m_i \mid x_i^o, x_i^m)] \Big\}. \end{aligned}$$

The expectation is taken with respect to the distribution of the missing covariates conditional on the observables,

$$\begin{aligned} f_\gamma(x_i^m \mid y_i, x_i^o, m_i) &= \frac{f_\gamma(y_i, x_i^o, x_i^m, m_i)}{\int_{\mathcal{X}_m} f_\gamma(y_i, x_i^o, x_i^m, m_i)\, dx_i^m} \\ &= \frac{f_\theta(y_i \mid x_i^o, x_i^m) f_\xi(x_i^o, x_i^m) f_\psi(m_i \mid x_i^o, x_i^m)}{\int_{\mathcal{X}_m} f_\theta(y_i \mid x_i^o, x_i^m) f_\xi(x_i^o, x_i^m) f_\psi(m_i \mid x_i^o, x_i^m)\, dx_i^m}. \end{aligned}$$

Further work proved that maximising this expectation is equivalent to the brute-force maximisation of the log-likelihood of the observables. It remains computationally challenging, because we must

compute $f_\gamma(x_{i,\text{mis}} \mid y_i, x_{i,\text{obs}}, m_i)$, including the integral in the denominator when missing covariates are continuous.

The E-step and the M-step

The EM algorithm is an iterative two-step procedure. At iteration ℓ we repeat:

Algorithm 5 (EM). E-step. Given the current value of the parameters $\gamma^{(\ell-1)}$, define the expected value of the log-likelihood function

$$Q\big(\gamma \mid \gamma^{(\ell-1)}\big) = \sum_i \int_{\mathcal{X}_{\text{mis}}} f_{\gamma^{(\ell-1)}}(x_{i,\text{mis}} \mid y_i, x_{i,\text{obs}}, m_i)\, \log f_\gamma(y_i, x_i, m_i)\, dx_{i,\text{mis}}.$$

M-step. Find the value of γ which maximises $Q(\gamma \mid \gamma^{(\ell-1)})$,

$$\hat\gamma = \arg\max_\gamma Q\big(\gamma \mid \gamma^{(\ell-1)}\big),$$

and repeat until the parameter vector, or the observed-data log-likelihood, changes by less than ϵ.

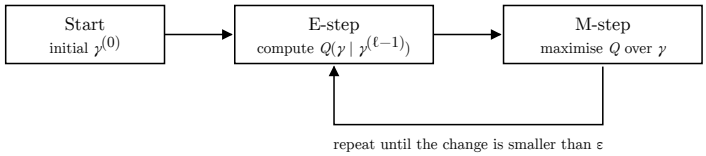


Figure 1. The EM iteration: the E-step fixes the distribution of the missing covariates at the current parameter value; the M-step updates the parameter.

Worked application: a salary survey

A salary survey records annual salary, years of experience and whether a respondent has a university degree. We analyse $\log(\text{salary})$, but the degree indicator is missing for some respondents due to a glitch in the IT infrastructure used to collect and store the data.

- **Observed:** $Y_i = \log(\text{annual salary}_i)$; Z_i: years of experience.
- **Partly observed:** $X_i = 1$ if the respondent has a university degree, 0 otherwise; X_i is missing for some respondents.
- **Target:** estimate the salary premium associated with a university degree, conditional on experience.

Step 1: specify the complete-data model

Regression model:

$$Y_i = \alpha + \beta X_i + \delta Z_i + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2),$$

$$f_\theta(y_i \mid x_i, z_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big\{ -\frac{1}{2}\Big(\frac{y_i - \alpha - \beta x_i - \delta z_i}{\sigma}\Big)^2 \Big\},$$

with $\theta = (\alpha, \beta, \delta, \sigma)$. Model for the discrete covariate subject to missingness (a model for the years of experience is not needed): $X_i \sim \text{Bernoulli}(\xi)$, $f_\xi(x_i) = \xi^{x_i}(1-\xi)^{1-x_i}$. Assuming independently distributed observations, the likelihood of θ and ξ given the data would be

$$L(\theta, \xi \mid y, x, z) = \prod_{i=1}^{n} f_\theta(y_i \mid x_i, z_i)\, \xi^{x_i} (1-\xi)^{1-x_i}.$$

Step 2: write the observed-data likelihood

Let $M_i = 1$ when the degree is observed and $M_i = 0$ when it is missing. Under MAR,

$$L_i(\theta, \xi) = \begin{cases} f_\theta(y_i \mid x_i, z_i)\, \xi^{x_i} (1-\xi)^{1-x_i}, & M_i = 1, \\ \displaystyle\sum_{x=0}^{1} f_\theta(y_i \mid x, z_i)\, \xi^{x} (1-\xi)^{1-x}, & M_i = 0. \end{cases}$$

A unit whose degree is observed contributes its recorded value. A respondent with a missing degree contributes both possible degree values, weighted by their probabilities: the missing respondent *remains in the likelihood*. Then $(\hat\theta, \hat\xi) = \arg\max_{\theta,\xi} \sum_{i=1}^{n} \log L_i(\theta, \xi)$.

Step 3: Bayes' rule for a missing degree

For a respondent whose degree is missing, define $w_i = P(X_i = 1 \mid Y_i = y_i, Z_i = z_i)$. Bayes' rule gives

$$w_i = \frac{\xi\, f_\theta(y_i \mid x = 1, z_i)}{\xi\, f_\theta(y_i \mid x = 1, z_i) + (1-\xi)\, f_\theta(y_i \mid x = 0, z_i)}.$$

The observed salary and experience update our belief about the missing degree. If the degree is observed there is no need to compute it; if it is missing, w_i is the probability that the ith unit has a degree, given the information on salary and years of experience. For example: what is the probability that a unit has a degree, given a salary of 100,000 euros and 20 years of experience?

Step 4: the E-step and the M-step

E-step. With $w_i^{\gamma^{(\ell-1)}}$ computed at the current parameter value,

$$\begin{aligned} Q\big(\gamma \mid \gamma^{(\ell-1)}\big) = & \sum_{i: m_i = 1} \Big[\log f_\theta(y_i \mid x_i, z_i) + x_i \log \xi + (1 - x_i) \log(1-\xi) \Big] \\ & + \sum_{i: m_i = 0} w_i \{ \log f_\theta(y_i \mid x = 1, z_i) + \log \xi \} \\ & + \sum_{i: m_i = 0} (1 - w_i) \{ \log f_\theta(y_i \mid x = 0, z_i) + \log(1-\xi) \}. \end{aligned}$$

M-step. Find γ which maximises $Q(\gamma \mid \gamma^{(\ell-1)})$. Repeat the two steps until the parameter estimates change negligibly. The R implementation below creates the two weighted possibilities ($x = 0$ with weight $1 - w_i$, $x = 1$ with weight w_i) for every respondent.

Note (added). The slide prints the observed-unit term as $\xi^{x_i}(1-x_i)^{1-x_i}$; the Bernoulli factor is $\xi^{x_i}(1-\xi)^{1-x_i}$, as written above.

R application: simulated salary survey

The synthetic data

We generate $n = 800$ synthetic respondents. Because the data-generating values are known, we can compare the complete-case and EM estimates with a benchmark.

$$Z_i = \max\{0, N(10, 5^2)\}, \qquad X_i \sim \text{Bernoulli}(0.55),$$

$$Y_i = 10.2 + 0.06Z_i + 0.25X_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 0.35^2).$$

The degree is then made incomplete: $p_i = \text{logit}^{-1}\{0.8 + 0.08(Z_i - \bar{Z})\}$; $Y_i = \log(\text{salary}_i)$ and Z_i remain observed; X_i is set to missing when a Bernoulli(p_i) answer is 0. Missingness depends on observed experience, so it is MAR. The next code generates the complete data, applies MAR missingness and reports the missing-degree rate.

```
set.seed(2026)                                  # Fix the random-number stream so this simulated sample can be
reproduced.
n <- 800                                        # Set the number of respondents.
experience <- pmax(0, rnorm(n, 10, 5))          # Draw experience; pmax() truncates below zero.
degree.true <- rbinom(n, 1, .55)                # Draw a 0/1 degree indicator with probability .55.
log_salary <- 10.2 + .06*experience + .25*degree.true + rnorm(n, 0, .35)   # Generate log salary with normal error.
salary.survey <- data.frame(log_salary, experience, degree = degree.true)  # Store variables by respondent.
p.answer <- plogis(.8 + .08*(experience - mean(experience)))   # Convert experience into a response probability.
salary.survey$degree_answered <- rbinom(n, 1, p.answer)       # Record 1 for an answer and 0 otherwise.
salary.survey$degree[salary.survey$degree_answered == 0] <- NA_integer_   # Mark unanswered degree values as missing.
mean(is.na(salary.survey$degree))               # Calculate the fraction of missing degree values.
with(salary.survey, tapply(log_salary, is.na(degree), mean))   # Compare mean log salary by missingness status.
```

```
> mean(is.na(salary.survey$degree))
[1] 0.30625
> with(salary.survey, tapply(log_salary, is.na(degree), mean))
   FALSE     TRUE
10.95702 10.88966
```

245 of 800 degree values are missing; 555 are recorded. `FALSE` means the degree is recorded, `TRUE` that it is missing. The group means describe the sample; they do not test whether MAR is true in real survey data.

EM for a binary missing covariate

```
em.degree <- function(y, z, x, maxit = 1000, tol = 1e-8) {   # EM for log salary with a missing binary covariate.
  obs <- !is.na(x)                              # Mark respondents whose degree value is recorded.
  pi <- mean(x[obs])                            # Start with the observed degree proportion.
  fit <- lm(y ~ x + z, subset = obs)            # Initialise regression using complete cases.
  beta <- coef(fit); sigma2 <- mean(resid(fit)^2)   # Save coefficients and residual variance.
  for (iter in seq_len(maxit)) {                # Repeat E- and M-steps up to maxit times.
    mu0 <- beta[1] + beta["z"] * z              # Predict log salary for degree = 0.
    mu1 <- mu0 + beta["x"]                      # Predict log salary for degree = 1.
    r0 <- dnorm(y, mu0, sqrt(sigma2)) * (1 - pi)    # Density times prior probability for degree = 0.
    r1 <- dnorm(y, mu1, sqrt(sigma2)) * pi          # Density times prior probability for degree = 1.
    w <- ifelse(obs, x, r1 / (r0 + r1))         # Use observed x or its posterior probability when missing.
    pseudo <- rbind(data.frame(y=y, x=0, z=z, wt=1-w),
                    data.frame(y=y, x=1, z=z, wt=w))   # Represent both possible x values with weights.
    fit.new <- lm(y ~ x + z, data=pseudo, weights=wt)  # Refit regression to expected complete data.
    beta.new <- coef(fit.new)                   # Update the regression coefficients.
    sigma2.new <- with(pseudo, weighted.mean((y - predict(fit.new))^2, wt))  # Update the weighted residual variance.
    pi.new <- mean(w)                           # Update the estimated degree proportion.
    if (max(abs(beta.new-beta), abs(sigma2.new-sigma2), abs(pi.new-pi)) < tol) break   # Stop when all updates are small.
    beta <- beta.new; sigma2 <- sigma2.new; pi <- pi.new   # Carry estimates into the next iteration.
  }
  list(beta=beta.new, sigma2=sigma2.new, pi=pi.new, posterior=w, iterations=iter)   # Return estimates and missing-x
probabilities.
}
em.fit <- em.degree(salary.survey$log_salary, salary.survey$experience, salary.survey$degree)   # Fit EM to the survey
data.
```

E-step: compute the posterior degree probabilities. M-step: update π, the regression coefficients and σ^2.

Comparing three analyses

```
cc.fit <- lm(log_salary ~ degree + experience, data = subset(salary.survey, !is.na(degree)))   # Full model only where
degree is observed.
omit.fit <- lm(log_salary ~ experience, data = salary.survey)   # Reduced model using all respondents.
em.beta <- setNames(unname(em.fit$beta), c("(Intercept)", "degree", "experience"))   # Align EM coefficient names with
lm().
rbind(true = c(`(Intercept)` = 10.2, degree = .25, experience = .06),
      complete_cases = coef(cc.fit), EM = em.beta)              # Compare estimates with known truth.
coef(summary(cc.fit))[, c("Estimate", "Std. Error")]            # Complete-case estimates and standard errors.
coef(summary(omit.fit))                                         # Estimates and standard errors after omitting degree.
```

```
c(n_total = nrow(salary.survey), complete_case_n = nobs(cc.fit),
  missing_degree = mean(is.na(salary.survey$degree)))          # Report sample counts and missingness.
```

The observed-data log-likelihood and EM standard errors

```
# Return the log-likelihood of the values actually observed.
obs.loglik <- function(par, y, z, x) {      # par: three coefficients, log variance and logit prevalence.
  b0 <- par[1]; bx <- par[2]; bz <- par[3]  # Unpack intercept, degree effect and experience effect.
  sigma <- exp(par[4] / 2)                  # Convert log variance to a positive standard deviation.
  pi <- plogis(par[5])                      # Convert logit prevalence to a probability between 0 and 1.
  mu0 <- b0 + bz*z                          # Conditional mean of log salary when degree = 0.
  mu1 <- mu0 + bx                           # Conditional mean when degree = 1.
  observed <- !is.na(x)                     # Identify records with a recorded degree value.
  ll <- numeric(length(y))                  # Space for one log-likelihood term per respondent.
  ll[observed] <- dnorm(y[observed], ifelse(x[observed] == 1, mu1[observed], mu0[observed]), sigma, log=TRUE) +
                  dbinom(x[observed], 1, pi, log=TRUE)       # Combine salary and degree probabilities.
  a <- dnorm(y[!observed], mu0[!observed], sigma, log=TRUE) + log1p(-pi)   # Contribution if the missing degree were 0.
  b <- dnorm(y[!observed], mu1[!observed], sigma, log=TRUE) + log(pi)      # Contribution if the missing degree were 1.
  ll[!observed] <- pmax(a,b) + log(exp(a-pmax(a,b)) + exp(b-pmax(a,b)))    # Log-sum-exp over both possible degrees.
  sum(ll)                                   # Add respondent contributions.
}

# Start at the EM estimates; log and logit transforms keep parameters unconstrained.
start <- c(unname(em.fit$beta), log(em.fit$sigma2), qlogis(em.fit$pi))
ll.em <- obs.loglik(start, salary.survey$log_salary, salary.survey$experience, salary.survey$degree)
ll.em                                       # Observed-data log-likelihood at the EM fit.
info <- optim(start, function(par) -obs.loglik(par, salary.survey$log_salary,
              salary.survey$experience, salary.survey$degree),
              method="BFGS", hessian=TRUE)$hessian   # Minimise the negative log-likelihood and compute its Hessian.
V.em <- solve(info)                          # Invert observed information to estimate the covariance.
se.em <- sqrt(diag(V.em))[1:3]               # Standard errors of the three regression coefficients.
se.cc <- coef(summary(cc.fit))[c("(Intercept)","degree","experience"), "Std. Error"]   # Complete-case standard errors.
rbind(complete_cases=se.cc, EM=se.em)        # Compare standard errors from the two methods.
```

An observed degree contributes a Normal density and a Bernoulli probability; a missing degree contributes the *mixture* over $X_i = 0$ and $X_i = 1$, weighted by $1 - \pi$ and π. The standard errors come from the observed information,

$$\hat{V}(\hat{\vartheta}) \approx \big\{ -\nabla^2 \ell_o(\hat{\vartheta}) \big\}^{-1}.$$

The first three entries give the standard errors of the intercept, the degree premium and the experience coefficient. Under MAR, EM uses all 800 records, so its degree-premium standard error should generally be smaller than the complete-case value.

Results

Table 5. Estimates (standard errors) from the simulated survey. Complete cases use 555 respondents; EM uses all 800.

Parameter	Truth	Complete cases	EM
Intercept	10.200	10.174 (0.0368)	10.195 (0.0314)
Degree	0.250	0.273 (0.0285)	0.272 (0.0272)
Experience	0.060	0.0604 (0.00294)	0.0585 (0.00247)

EM converges in 11 iterations, with $\hat{\pi} = 0.5377$, $\hat{\sigma}^2 = 0.1108$ and observed-data log-likelihood -657.0386. Here EM gives modestly smaller model-based standard errors; the gain depends on the missingness pattern and the information in the observed outcomes.

Omitting the degree

Table 6. Model without degree, fitted to all 800 respondents.

	Estimate	Std. Error
(Intercept)	10.33540076	0.029459832
experience	0.05912084	0.002613758

The reduced model uses all 800 respondents. In this simulation degree and experience were generated independently, so omitting the degree mainly shifts the intercept, while the experience slope stays near 0.06. This is design-specific: if degree attainment and experience were associated, omitting the degree could also bias the estimated experience slope.

Why omitting the degree can create bias

The complete-data relationship is $Y_i = \alpha + \beta X_i + \gamma Z_i + \varepsilon_i$ with X_i the degree indicator and Z_i experience. If X_i is omitted, the conditional mean becomes

$$\mathbb{E}[Y_i \mid Z_i] = \alpha + \gamma Z_i + \beta\, \mathbb{E}[X_i \mid Z_i].$$

The degree premium β is no longer estimated at all. If $\mathbb{E}[X_i \mid Z_i]$ varies with experience, the omitted term is absorbed into the fitted intercept and slope; therefore the coefficient of experience can be biased even though experience itself is fully observed. Here $X_i \perp Z_i$, so omitting the degree shifts the intercept by $\xi\beta \approx 0.55 \times 0.25 = 0.1375$ (compare $10.335 \approx 10.2 + 0.1375$ in the table above). If degree and experience were associated, the experience slope would also be biased.

A repeated-sample check: precision

The standard deviation of an estimator across repeated samples approximates its sampling variability.

```
one.sample <- function() {                  # Simulate one dataset and return both degree-effect estimates.
  n <- 800                                   # Respondents per simulated dataset.
  z <- pmax(0, rnorm(n, 10, 5))              # Simulate experience and truncate below zero.
  x <- rbinom(n, 1, .55)                     # Simulate the binary degree indicator.
  y <- 10.2 + .06*z + .25*x + rnorm(n, 0, .35)   # Generate log salary from the known model.
  p <- plogis(.8 + .08*(z - mean(z)))        # Response chance using observed experience.
  x.obs <- x                                 # Copy the complete degree values before introducing missingness.
  x.obs[rbinom(n, 1, p) == 0] <- NA_integer_  # Apply MAR missingness to degree.
  d <- data.frame(y, z, x = x.obs)           # Store one simulated survey.
  cc <- lm(y ~ x + z, data = d)              # Estimate from complete cases.
  em <- em.degree(y, z, x.obs)               # Estimate with EM using all records.
  c(complete_cases = coef(cc)["x"], EM = em$beta["x"])   # Return degree effects only.
}
set.seed(2027)                               # Fix the random stream for the repeated-sample check.
sampling.dist <- t(replicate(1000, one.sample()))   # 1,000 independent datasets; transpose the estimates.
apply(sampling.dist, 2, sd)                  # Empirical SDs of each method's estimates.
```

```
             complete_cases.x       EM.x
Empirical SD       0.02983129 0.02900454
```

Across these 1,000 samples the EM degree estimates have about 2.8% lower empirical standard deviation. This is a modest gain for this data-generating setup, not a universal percentage. (A lighter version of the check on the slides uses `replicate(100, one.sample())`; the complete-case output is typically the larger one, because the same degree premium is estimated from fewer observed degree values.)

Results: three analyses, three lessons

Table 7. Comparison of the three analyses of the salary survey.

Analysis	Information used	What to see
EM	All 800 respondents; the missing degree is represented by w_i.	The degree premium is close to the true value $\beta = 0.25$ and the available information is used efficiently.
Complete cases	Only respondents whose degree is recorded.	Under MAR the estimate is still centred near 0.25, but the standard error is larger because observations have been discarded.
Omit degree	All respondents, but fit $Y \sim Z$.	The degree premium cannot be estimated; the intercept absorbs the average degree effect. If degree and experience are associated, the experience coefficient is biased as well.

Main empirical pattern: under MAR, EM and complete-case analysis target the same degree premium, but complete cases have larger uncertainty. Omitting the degree changes the model and may bias the remaining coefficients.

Recap, take-home messages and references

The analysis workflow

1. Define the target population and the quantity of interest.
2. Carefully analyse the data-extraction process to ensure the data are devoid of any flaw for use in a statistical analysis.
3. Describe the data and investigate why values are missing.
4. Based on 3, state the missing-data mechanism: MCAR, MAR or MNAR.
5. Use the observed-data likelihood; estimate the parameters and the uncertainty associated with their estimation.

A useful discipline: start with the data collection and the analysis of its potential flaws; then choose the inferential strategy and the corresponding computational method.

Take-home messages

1. Missingness is part of the data-generating process.
2. Complete cases are not automatically representative of the target population.
3. MCAR, MAR and MNAR are assumptions about *why* values are absent.
4. Under MAR, likelihood-based methods can use incomplete records without filling them in first.
5. EM is a computational route to maximum likelihood.
6. When MNAR is plausible, report sensitivity to the missingness assumption.

References (Part I)

- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581–592.
- Little, R. J. A. and Rubin, D. B. (2019). *Statistical Analysis with Missing Data*, 3rd edition. Wiley.
- Dempster, A. P., Laird, N. M. and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–22.

- Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83(404), 1198–1202.
- Guindani, M. Handling missing data. Lecture 11, *Statistical Methods for Correlated Data*.

Part II

Cross-sectional and Longitudinal Data Analysis with Latent Variable Models

Lecture 2 of 3 · F. Ungolo, UNSW Sydney

Background and learning objectives

The analysis of rectangular datasets (implicitly) assumes that individual observations are independent of one another (e.g. voting preference in the 2026 elections, log-salary in 2026): each record corresponds to a single unit in the sample. The interest is now in the statistical analysis of individual measurements *over time* for a sample of units (the voting preferences over the last 20 years, log-salary over a career, blood pressure) and in their determinants.

Longitudinal versus time-series data. Time-series data are concerned with the statistical behaviour of one (or several) variable(s) over time and its relationship with its past values (e.g. the quarterly unemployment rate, the yearly inflation rate). Longitudinal data analysis is concerned with repeated measurements of *multiple individuals*.

A voting survey over elections

Table 8. Each row is one respondent–election record; the same people are followed across successive elections.

Voter ID	Election	Age	Income band	Main policy concern	Vote choice
101	2016	25	Medium	Employment	Party A
101	2020	29	Medium	Housing	Party B
101	2024	33	High	Housing	Party B
102	2016	42	Low	Taxes	Party B
102	2020	46	Low	Public services	Party B
102	2024	50	Medium	Public services	Party C
103	2016	43	High	Climate	Party A
103	2020	47	High	Climate	Party A
103	2024	51	High	Taxes	Party B

Questions: What determines the choice of the party? Do respondents change their vote choice across elections? How are income and policy concerns related to vote choice? Do choices vary between people as well as within the same person over time?

Learning objectives

- Longitudinal versus cross-sectional data;
- dependence between repeated measurements of the same unit;
- linear mixed models;
- statistical inference and prediction for linear mixed models;
- model checking;
- introduction to generalised linear and nonlinear mixed models;
- application in R.

Running example: student grades and study hours

A sample of 16 students is followed for four semesters. At each semester the dataset records the course grade and the average daily study hours. The teaching dataset contains $16 \times 4 = 64$ student–semester records; the table below is an excerpt. Is study time associated with grades? Do grades change over semesters? Do students differ in starting grades and progress?

Table 9. Excerpt of the student panel (students 1 and 2).

Student	Semester	Study hours/day	Grade
1	1	0.8	40
1	2	0.9	39
1	3	1.0	38
1	4	1.1	36
2	1	1.2	48
2	2	1.3	47
2	3	1.4	47
2	4	1.5	46

A first model: pooled linear regression

For student $i = 1, \ldots, 16$ and semester $t = 1, \ldots, 4$ start with a linear regression,

$$Y_{i,t} = \beta_0 + \beta_1 X_{i,t} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2),$$

where $Y_{i,t}$ is the average grade in semester t for student i and $X_{i,t}$ the corresponding average study hours per day. One line is fitted to all 64 student–semester records; β_0 measures the average grade across all students; β_1 summarises the change in the average grade for a change in the average study hours per day; the 64 error terms $\varepsilon_{i,t}$ are independently distributed. *Question: are four grades from one student as independent as grades from four different students?*

The scatterplot shows the 64 student–semester records and the pooled linear fit (Figure 2).

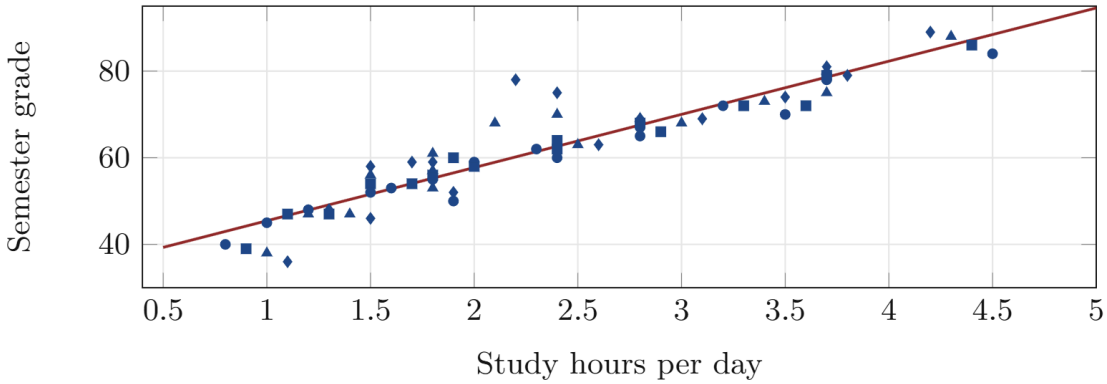


Figure 2. Study hours per day versus semester grade. Each point is one student–semester record; circle, square, triangle and diamond markers denote semesters 1, 2, 3 and 4. The red line is the pooled linear fit.

R: creating the artificial dataset and fitting the linear model

```
# Enter each student's grades in semester order.
grades <- matrix(c(
  40,39,38,36,  48,47,47,46,  53,54,53,52,  59,60,61,59,
  62,62,63,63,  67,68,69,69,  72,72,73,74,  78,79,80,81,
```

```
  45,47,47,48,  55,56,57,59,  65,66,68,69,  84,86,88,89,
  52,54,56,58,  70,72,75,79,  60,64,70,75,  50,58,68,78
), nrow = 16, byrow = TRUE)                 # One row per student.
# Record semester-1 hours and each student's change in hours.
hours_start  <- c(.8,1.2,1.6,2.0,2.3,2.8,3.2,3.7,
                  1.0,1.8,2.8,4.5,1.5,3.5,2.4,1.9)
hours_change <- c(.1,.1,.1,-.1,.1,0,.1,0,
                  .1,0,.1,-.1,0,.1,0,.1)
# Create the semester labels and elapsed-time values.
semester <- rep(1:4, times = nrow(grades))
time <- semester - 1
# Repeat student IDs and calculate hours in each semester.
id <- factor(rep(1:nrow(grades), each = 4))
study_hours <- round(rep(hours_start, each = 4) +
                     rep(hours_change, each = 4) * time, 1)
# Combine the variables in one row-per-student-semester dataset.
student_data <- data.frame(
  id = id, semester = semester, study_hours = study_hours,
  grade = as.vector(t(grades))
)
head(student_data)                          # Inspect the first few student-semester records.

# Fit one line to all student-semester records.
fit_lm <- lm(grade ~ study_hours, data = student_data)
summary(fit_lm)       # Coefficient estimates, standard errors, residual summaries.

# Choose a different plotting symbol for each semester and plot the pooled line.
semester_pch <- c(16, 15, 17, 18)
plot(student_data$study_hours, student_data$grade,
     pch = semester_pch[student_data$semester], col = "steelblue",
     xlab = "Study hours per day", ylab = "Semester grade")
abline(fit_lm, col = "firebrick", lwd = 2)   # Add the pooled regression line.
legend("topleft", paste("Semester", 1:4),
       pch = semester_pch, col = "steelblue", bty = "n")
```

The model is a useful starting point. Its usual standard errors rely on an independence assumption that this panel does not satisfy.

Limitations of the simple linear regression

The simple linear model assumes that all grades, given the study hours, are independently distributed: $\mathrm{Cov}(\varepsilon_{i,t}, \varepsilon_{i,s}) = 0$ for $t \neq s$. But:

- a student's individual characteristics, such as background preparation or talent, can affect several semesters, so their grades tend to move together;
- a straight regression line does not capture the effect of a common time trend or different student trajectories.

Treating each observation as independent can make the standard errors of the parameters too small, hence the confidence intervals too narrow. *Is it the end of a linear model?* Use the simple linear fit as a starting point to identify a specific pattern in the data. Another useful graphical tool is the plot of each student's trajectory, which guides a model for time and student-level variation.

Individual grades are correlated

Table 10. Estimated linear correlations between semesters across the 16 students of the teaching panel.

	Sem. 1	Sem. 2	Sem. 3	Sem. 4
Sem. 1	1.000	0.985	0.933	0.851
Sem. 2	0.985	1.000	0.979	0.926
Sem. 3	0.933	0.979	1.000	0.982
Sem. 4	0.851	0.926	0.982	1.000

Grades from the same student are strongly associated, even four semesters apart. These are correlations between repeated outcomes, not evidence that one semester causes another. A linear mixed model can capture this dependence through shared student effects. In R: `round(cor(grades), 3)`.

Spaghetti plot. Plotting the grades of each student over the four semesters (and the average grade) shows that students start at different grade levels, that their paths are not parallel (semester-to-semester change varies) and that the average line hides these differences; random intercepts and slopes can represent them.

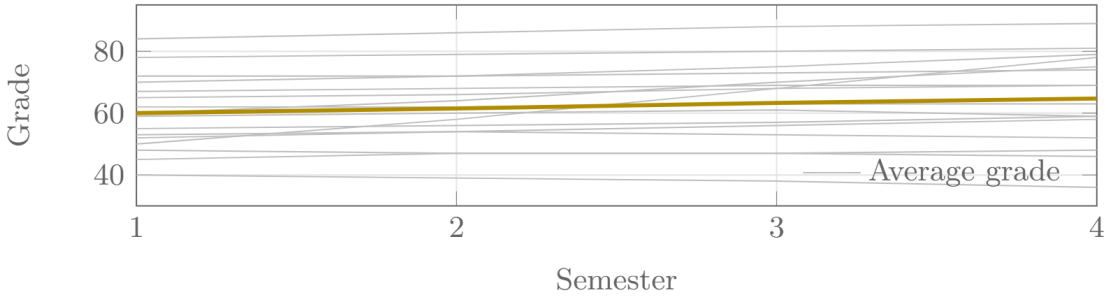


Figure 3. Spaghetti plot: grades of each student over the four semesters (thin grey lines) and the average grade (thick line). Students start at different levels and their paths are not parallel.

A first extension: a semester-trend variable

Let $u = t - 1$ be the elapsed semesters and $X^c_{i,t} = X_{i,t} - 3$ the study hours centred at 3 per day:

$$Y_{i,t} = \beta_0 + \beta_1 X^c_{i,t} + \beta_2 u + \beta_3 u X^c_{i,t} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{\text{iid}}{\sim} N(0, \sigma^2_\varepsilon).$$

β_0: expected semester-1 grade at 3 study hours per day; β_1: association between study hours and grade in semester 1; β_2: average grade change per semester at 3 study hours per day; β_3: how that association changes with each semester. Still limited: every student has the same starting grade and time trend, and the errors are assumed independent.

The linear mixed-effects model

Definition 6 (linear mixed-effects model).

$$Y_{i,t} = \beta^\top x_{i,t} + b_i^\top z_{i,t} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{\text{iid}}{\sim} N(0, \sigma^2_\varepsilon), \qquad b_i \overset{\text{iid}}{\sim} N(0, D).$$

Here $x_{i,t} = (x_{1,i,t}, \ldots, x_{p,i,t})$ and $z_{i,t} = (z_{1,i,t}, \ldots, z_{q,i,t})$ are vectors of covariates (the two vectors can include similar covariates); $\beta = (\beta_1, \ldots, \beta_p)$ is the vector of *fixed-effect* parameters; b_i is a q-dimensional vector of unobserved *random-effect* parameters (independent of $\varepsilon_{i,t}$) capturing the differing effect of the covariates $z_{i,t}$; D describes the variation between students and $\varepsilon_{i,t}$ the remaining within-student variability.

Remark. The random effects are individual-specific. For the student grades this induces dependence between a student's repeated observations.

Key features

- Linear mixed models allow one curve to be estimated for each individual: the individual random effect captures the different intercept across units, and the different sensitivity of units to the value of a covariate. This way we model both the variability across units and within units.

- We can still estimate the population-wide effect of each covariate $x_{i,t}$ (the *marginal model*):

$$\mathbb{E}[Y_{i,t} \mid x_{i,t}, z_{i,t}] = \mathbb{E}(\beta x_{i,t} + b_i z_{i,t} + \varepsilon_{i,t}) = \beta x_{i,t} + \mathbb{E}(b_i) z_{i,t} + \mathbb{E}(\varepsilon_{i,t}) = \beta x_{i,t}$$

  (this result is possible for linear models only).

Mean and covariance of the repeated measurements

Vectorising the repeated measurements of the ith unit as $Y_i = (Y_{i,1}, \dots, Y_{i,T})$,

$$Y_i = x_i \beta + z_i b_i + \varepsilon_i.$$

If $b_i \sim N(0, D)$ and $\varepsilon_i \sim N(0, R_i)$, with b_i independent of $\varepsilon_i = (\varepsilon_{i,1}, \dots, \varepsilon_{i,T})$, then

$$\mathbb{E}(Y_i) = x_i \beta, \qquad \mathrm{Var}(Y_i) = z_i D z_i^\top + \sigma_\varepsilon^2 I.$$

$x_i \beta$ gives the population mean of Y_i over time; $z_i D z_i^\top$ is the covariance of the measurements for the ith unit; $\sigma_\varepsilon^2 I$ is the remaining variability. The model describes both the average trajectory and how grades from the same unit co-vary.

The role of the latent variable

In the *mixed-effects model*, b_i captures the unit-specific effect of a certain covariate not explained by the fixed part of the model; in the student example it is estimated through the repeated measurements. A random effect is a model component for unobserved heterogeneity and within-student dependence.

In a general regression model for *cross-sectional* data the analogue is an unobserved component that affects the outcome:

$$Y_i = \beta x_i + \gamma c_i + \varepsilon_i, \qquad c_i \sim f_\tau(\cdot).$$

For example, if Y_i is the logarithm of the future human lifespan, we study its association with observable variables x_i (age, sex, ...) and unobservable features c_i such as health status and wealth. In both cases the latent quantity is inferred from the observed data through a statistical model.

The model for the students' grades

Let $t = 1, 2, 3, 4$ label the semester, $u = t - 1$ the elapsed semesters and $X^c_{i,t} = X_{i,t} - 3$:

$$Y_{i,t} = \underbrace{\beta_0 + \beta_1 u + \beta_2 X^c_{i,t} + \beta_3 u X^c_{i,t}}_{\text{population (fixed) part}} + \underbrace{b_{0,i} + b_{1,i} u}_{\text{student-specific (random) part}} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2).$$

- β_0, β_1: average semester-1 grade and semester-to-semester change at 3 study hours per day;
- β_2, β_3: association of study time with grade in semester 1 and how that association changes over semesters;
- $b_i = (b_{0,i}, b_{1,i})^\top \sim N(0, D)$: student i's deviation from the overall semester-1 grade and from the overall time change, i.e. the student-specific starting grade and trajectory;
- $\varepsilon_{i,t}$: residual accounting for the variation not explained by the regression part, with $b_i \perp \varepsilon_{i,t}$; it captures the within-student variability.

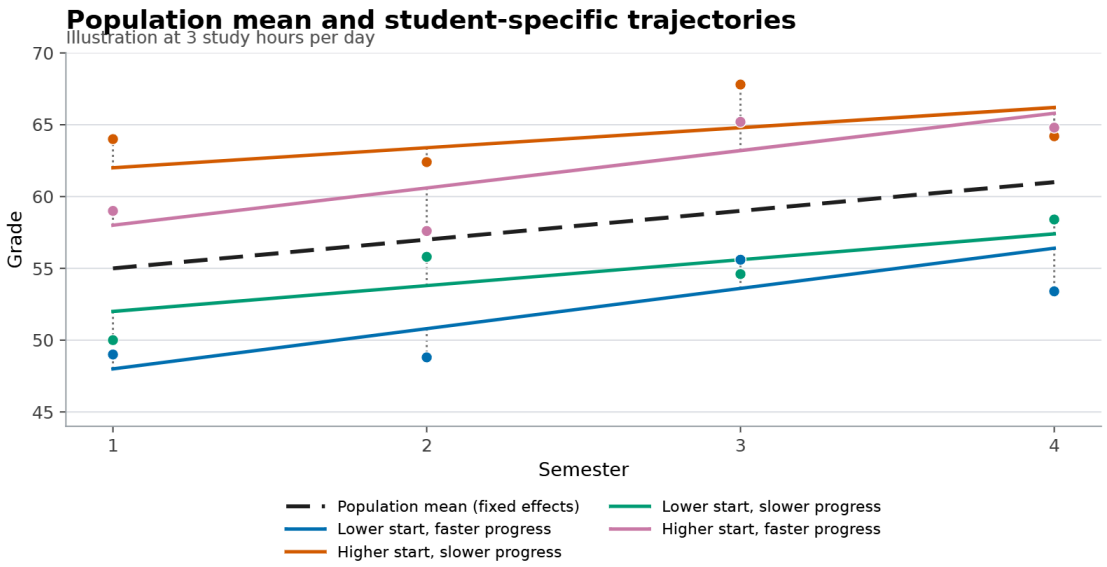


Figure 4. What the two levels look like, at 3 study hours per day. Dashed black: fixed-effects (population) mean. Coloured lines: mean plus student effects (lower or higher start, faster or slower progress). Points: grades; dotted segments: residuals. Component values are illustrative.

What can be learnt from the model?

- **Average baseline and trajectory:** the expected semester-1 grade and the average change per semester (β_0 and β_1).
- **Study-hours association:** the grade difference per additional study hour per day (β_2) and whether that association changes over semesters (β_3).
- **Student heterogeneity:** how much starting grades and rates of change vary between students ($b_{0,i}$ and $b_{1,i}$).
- **Student-level prediction:** a future grade for a new student or for a student already observed.

Fixed-effect estimates address average associations; variance components describe heterogeneity; predictions combine the two.

Random intercept and random slope

Consider again $Y_{i,t} = \beta_0 + \beta_1 u + \beta_2 X^c_{i,t} + \beta_3 u X^c_{i,t} + b_{0,i} + b_{1,i} u + \varepsilon_{i,t}$.

- **Random intercept.** β_0 is the expected semester-1 grade for a student at the reference study time; $b_{0,i}$ is student i's departure from that average starting grade. The same $b_{0,i}$ appears in every semester, shifting the student's trajectory up or down. Two students with the same measured covariates can therefore have different predicted grades.
- **Random slope.** β_1 is the average grade change per semester at the reference study time; $b_{1,i}$ is how student i's change differs from that average. Student i's time slope is $\beta_1 + b_{1,i}$. The random intercept shifts the starting point; the random slope changes the tilt of the trajectory. Random slopes are useful when the spaghetti plot shows students changing at different rates.

How starting grades and progress vary

$$\begin{pmatrix} b_{0,i} \\ b_{1,i} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, D\right], \qquad D = \begin{pmatrix} \sigma_0^2 & \sigma_{0,1} \\ \sigma_{0,1} & \sigma_1^2 \end{pmatrix}, \qquad \rho_{0,1} = \frac{\sigma_{0,1}}{\sigma_0 \sigma_1}.$$

σ_0: between-student spread in semester-1 grades; σ_1: between-student spread in grade change per semester; $\rho_{0,1} > 0$: students who start higher tend, on average, to improve faster, while $\rho_{0,1} < 0$

indicates the opposite pattern. These describe association in the model; they do not, by themselves, explain *why* students differ.

Likelihood inference for the linear mixed model

The collected data (y, x, z) are used to estimate the unknown parameters β, D and σ_ε^2. The likelihood is

$$L(\beta, D, \sigma_\varepsilon^2) = \prod_{i=1}^{N} \int_{\mathcal{B}_i} f_{\beta,\sigma_\varepsilon^2}(y_i \mid x_i, z_i, b_i)\, f_D(b_i)\, db_i, \qquad f_{\beta,\sigma_\varepsilon^2}(y_i \mid x_i, z_i, b_i) = \prod_{t=1}^{T} f_{\beta,\sigma_\varepsilon^2}(y_{i,t} \mid x_{i,t}, z_{i,t}, b_i).$$

For Normally distributed $y_{i,t}$ and b_i the integral is available in closed form: $\int f(y_i \mid x_i, z_i, b_i) f_D(b_i)\, db_i$ is the density of a Normal distribution with mean $x_i^\top \beta$ and variance $V_i = z_i^\top D z_i + \sigma_\varepsilon^2 I$. Optimising the log-likelihood (omitting multiplicative constants) is more efficient:

$$\log L(\beta, D, \sigma_\varepsilon^2) = -\frac{1}{2} \sum_{i=1}^{n} \Big[\log |V_i| + (y_i - x_i^\top \beta)^\top V_i^{-1} (y_i - x_i^\top \beta) \Big],$$

the log-density of the multivariate normal distribution. Given the estimates of σ_ε^2 and D, $\hat\beta$ is available in closed form,

$$\hat\beta = \Big(\sum_{i=1}^{n} x_i^\top V_i x_i \Big)^{-1} \Big(\sum_{i=1}^{n} x_i^\top V_i y_i \Big),$$

and meets the asymptotic properties of the maximum-likelihood estimator (consistency and asymptotic normality). (As printed on the slide; the generalised-least-squares form uses V_i^{-1}, as in Part III.)

Application: a panel of 800 students

A sample of 800 students is followed over four semesters: 3,200 potential student–semester grade records. Each student has a baseline average of daily study hours, centred at 3 hours per day. Grades depend on semester, study hours and student-specific starting grades and progress.

Define students and predictors; generate complete grade histories

```
set.seed(2026)                    # Fix the random-number stream so the sample can be reproduced.
n <- 800                          # Number of students.
semester <- 1:4
time <- 0:3                       # Elapsed semesters; semester 1 is the reference.
# Generate baseline study hours and centre them at 3 hours per day.
study_hours <- pmin(7, pmax(0, rnorm(n, mean = 3, sd = 1)))
study_hours_c <- study_hours - 3
# Generate student-specific starting grades and time slopes.
b0 <- rnorm(n, mean = 0, sd = 5)
b1 <- rnorm(n, mean = 0, sd = 0.6)
# Start a wide dataset with one row per student.
wide <- data.frame(id = factor(seq_len(n)), study_hours_c = study_hours_c)
# Generate one grade at each semester.
for (j in seq_along(semester)) {
  # Add the population mean and student-specific deviations.
  mean_grade <- 55 + 2 * time[j] + 4 * study_hours_c +
                0.3 * time[j] * study_hours_c + b0 + b1 * time[j]
  # Add occasion-level error and save that semester's grades.
  wide[[paste0("grade", semester[j])]] <- mean_grade + rnorm(n, mean = 0, sd = 5)
}
```

The data-generating model has $\beta = (55,\ 2,\ 4,\ 0.3)$ for intercept, time, centred hours and their interaction, $b_{0,i} \sim N(0, 5^2)$, $b_{1,i} \sim N(0, 0.6^2)$ and $\varepsilon_{i,t} \sim N(0, 5^2)$.

Reshape and inspect the panel; fit the mixed model

```
library(tidyr); library(ggplot2)           # Tools for reshaping and plotting.
# Convert grade1--grade4 columns to one row per student-semester.
long <- pivot_longer(wide, cols = starts_with("grade"),
                     names_to = "occasion", values_to = "grade")
long$semester <- as.integer(sub("grade", "", long$occasion))   # Extract the semester number.
long$time <- long$semester - 1                                  # Define elapsed time.
long$id <- factor(long$id)                                      # Grouping factor for lme4.
# Plot each student's grades and the mean at each semester.
ggplot(long, aes(semester, grade, group = id)) +
  geom_line(alpha = .08) +
  stat_summary(aes(group = 1), fun = mean, geom = "line",
               colour = "red", linewidth = 1)

library(lme4)                               # Package for linear mixed-effects models.
# Fit the population trend and student-specific intercept/slope.
mixed_fit <- lmer(grade ~ time * study_hours_c + (time | id),
                  data = long, REML = FALSE)
summary(mixed_fit)       # Fixed effects, random effects, and residual variation.
```

Thin lines show individual grades; the red line is the semester mean.

Reading the `lmer` output and formula

- **Fixed-effects table:** estimates, standard errors and t values for the average trajectory and study-hours associations. `(Intercept)`: expected semester-1 grade at 3 study hours per day. `time`: average grade change per semester at 3 study hours per day. `study_hours_c`: grade association per additional hour per day in semester 1. `time:study_hours_c`: how that association changes each semester.
- **Random-effects table:** student-level standard deviations and their correlation; `Residual` is the occasion-level variation.
- For Gaussian `lmer` fits, `summary()` does not provide fixed-effect p values by default; use confidence intervals or a justified model comparison.

In the formula `grade ~ time * study_hours_c + (time | id)`: `time * study_hours_c` includes time, baseline study hours and their interaction as population effects; `(time | id)` gives each student a random intercept and a random time slope, with an estimated covariance; the grouping factor `id` tells `lme4` which records belong to the same student.

Model comparison

Comparing fixed effects with maximum likelihood

`anova(m0, m1)` compares these nested ML fits and reports a likelihood-ratio test (LRT). The LRT statistic is $2\{\ell(m_1) - \ell(m_0)\}$; under $H_0 : \beta_3 = 0$ it is approximately χ^2_1, and a small p-value supports the interaction. Use ML because REML likelihoods depend on the fixed-effects formula; compare models on the same data and random-effects structure. After selecting the fixed effects, refit by REML, which accounts for the estimation of the fixed effects and usually gives less biased variance-component estimates.

```
# Fit the model without a time-by-hours interaction using ML.
m0 <- lmer(grade ~ time + study_hours_c + (time | id), data = long, REML = FALSE)
# Fit the nested model with the interaction, also using ML.
m1 <- lmer(grade ~ time * study_hours_c + (time | id), data = long, REML = FALSE)
# anova() reports the LRT, the difference in degrees of freedom, and its p-value.
anova(m0, m1)
# Refit with REML for final variance-component estimates.
final_fit <- update(m1, REML = TRUE)
```

AIC: fit with a complexity penalty

The three models under analysis are ranked using information criteria such as the Akaike information criterion,

$$\mathrm{AIC} = -2\,\ell(\hat{\theta}) + 2k.$$

$\ell(\hat{\theta})$ is the maximised log-likelihood (better fit increases the log-likelihood and lowers AIC); k is the number of free parameters estimated from the data; the term $2k$ penalises models with more parameters. Among comparable candidate models, prefer the one with the lower AIC.

Counting parameters

The study-hours-only model has two fixed effects; the time-trend model and the LMM have four: $\beta_0, \beta_1, \beta_2, \beta_3$. The LMM covariance parameters are $\mathrm{Var}(b_{0i})$, $\mathrm{Var}(b_{1i})$ and $\mathrm{Cov}(b_{0i}, b_{1i})$. The student-specific random effects are integrated over their distribution; they are *not* counted as one extra parameter per student.

Table 11. Parameter count for the three models.

Model	Fixed	Random covariance	Residual variance	k
Study-hours-only linear model	2	0	1	3
Time-trend linear model	4	0	1	5
Random intercept and slope LMM	4	3	1	8

AIC comparison in R and results

```
# Fit all three models to the same complete panel, using maximum likelihood.
lm_hours <- lm(grade ~ study_hours_c, data = long)             # Study hours only.
lm_time  <- lm(grade ~ time * study_hours_c, data = long)       # Time-trend linear model.
mixed_fit_ml <- lmer(grade ~ time * study_hours_c + (time | id),
                     data = long, REML = FALSE)                # Student-specific intercepts and slopes.
# Compare AIC values and the difference of each model from the minimum.
aic_tab <- AIC(lm_hours, lm_time, mixed_fit_ml)
aic_tab$Delta_AIC <- aic_tab$AIC - min(aic_tab$AIC)
aic_tab
```

Table 12. AIC comparison on the same 3,200 complete grades from 800 students.

Model	k	AIC	ΔAIC
Study-hours-only linear model	3	21 983.96	1 189.32
Time-trend linear model	5	21 666.25	871.61
Random intercept and slope LMM	8	20 794.64	0.00

The time-trend model lowers AIC by 317.71 relative to the study-hours-only model. The LMM lowers AIC by a further 871.61 relative to the time-trend model, reflecting the value of modelling student-specific trajectories in this simulation.

Fixed-effect estimates and variance components

```
# Population-average coefficients and approximate 95% Wald intervals.
estimate <- fixef(final_fit)
ci_fixed <- confint(final_fit, parm = "beta_", method = "Wald", level = 0.95)
cbind(estimate = estimate, ci_fixed)

VarCorr(final_fit)     # Estimated student-level covariance matrix (SDs and correlation).
sigma(final_fit)       # Residual standard deviation.
```

A Wald 95% interval is estimate ± 1.96 standard errors under approximate normality. It summarises uncertainty in a fixed-effect coefficient; if it excludes zero, it gives approximate evidence of a nonzero association. `VarCorr()` reports the random-effect variances and covariance (its printout shows standard deviations and correlations); `sigma()` returns $\hat{\sigma}_\varepsilon$, and the residual variance is $\hat{\sigma}^2_\varepsilon$.

Model checking

The checks to carry out: Is the average grade trend approximately linear over four semesters? Are residuals associated with the fitted grade or the semester? Do residuals show remaining patterns within a student? Are the random-intercept and random-slope distributions plausible? Did the optimiser converge, and is the fitted model singular? Does prediction work for students not used in fitting?

```
# Conditional residuals against fitted grades, and a normal QQ plot.
diagnostic <- long
diagnostic$fitted <- predict(final_fit, newdata = diagnostic, re.form = NULL)   # Uses each student's random effects.
diagnostic$resid  <- diagnostic$grade - diagnostic$fitted
plot(diagnostic$fitted, diagnostic$resid,
     xlab = "Fitted grade", ylab = "Conditional residual")
abline(h = 0, lty = 2)
qqnorm(diagnostic$resid, xlab = "Theoretical normal quantiles",
       ylab = "Residual quantiles")
qqline(diagnostic$resid)

# Compare residual distributions at each semester.
boxplot(resid ~ factor(semester), data = diagnostic,
        xlab = "Semester", ylab = "Conditional residual")

# Inspect student-specific intercept and time-slope distributions.
re <- ranef(final_fit)$id
qqnorm(re[, "(Intercept)"]); qqline(re[, "(Intercept)"])
qqnorm(re[, "time"]);        qqline(re[, "time"])
# Check whether the fitted random-effects structure is singular.
isSingular(final_fit, tol = 1e-4)
```

- A pattern-free cloud around zero supports the mean structure; curvature can suggest a missing term. Points close to the reference line in the QQ plot are consistent with approximately normal residuals.
- Semester-specific shifts in the boxplots may point to a missing time effect or predictor; a larger spread at later semesters may indicate changing residual variance; repeated patterns within students may mean the time trend or dependence structure needs review. Patterns are diagnostic clues: investigate them before changing the model.
- The QQ plots of the estimated effects assess whether they look roughly normal. A *singular* fit warns that the data may not support the full random-effects structure.

Student-specific estimates, shrinkage and prediction

Estimating b_i and shrinkage

For a fitted Gaussian linear mixed model, student i's value of b_i is estimated by its conditional expectation,

$$\hat{b}_i = \mathbb{E}_{\hat{\theta}}(b_i \mid y_i, x_i, z_i) = \hat{D} z_i^\top \hat{V}_i^{-1}(y_i - x_i\hat{\beta}), \qquad \hat{V}_i = z_i \hat{D} z_i^\top + \hat{\sigma}_\varepsilon^2 I.$$

y_i is the vector of observed grades; x_i and z_i describe the fixed and random parts of the model; $\hat{\theta} = (\hat{\beta}, \hat{D}, \hat{\sigma}_\varepsilon^2)$ contains the fitted fixed effects and variance parameters; $\hat{b}_i$ is the expected student-specific departure given the observed grades and the fitted parameters. *Short or noisy histories are shrunk toward zero; clearer repeated patterns receive a larger adjustment.* In a Gaussian LMM this conditional mean equals the conditional mode returned by `ranef()`.

```
b_hat <- ranef(final_fit)$id     # Estimated random intercepts and time slopes per student.
head(b_hat)
```

Out-of-sample comparative performance

The predictive ability of the linear mixed model is compared with that of linear models. The models are fitted on the first three semesters, the grades of semester 4 are forecast, and the forecasts are compared with the actual observations:

$$\hat{Y}_{i,4} = x_{i,4}^\top \hat{\beta} + z_{i,4}^\top \hat{b}_i \ \text{(linear mixed model)}, \qquad \hat{Y}_{i,4} = x_{i,4}^\top \hat{\beta} \ \text{(linear model)}.$$

The accuracy is measured by the root mean squared error and the mean absolute error over the 800 students,

$$\text{RMSE} = \sqrt{\frac{1}{800}\sum_i (\hat{Y}_{i,4} - Y_{i,4})^2}, \qquad \text{MAE} = \frac{1}{800}\sum_i |\hat{Y}_{i,4} - Y_{i,4}|.$$

```
# Train on semesters 1--3 and reserve semester 4 for testing.
train <- subset(long, semester < 4)
test  <- subset(long, semester == 4)
hours_fit <- lm(grade ~ study_hours_c, data = train)             # Study-hours-only linear model.
time_fit  <- lm(grade ~ time * study_hours_c, data = train)      # Linear model with a time trend and interaction.
cv_fit <- lmer(grade ~ time * study_hours_c + (time | id),
               data = train, REML = TRUE)                        # Mixed model with student-specific effects.

# Predict semester-4 grades using each fitted model.
predictions <- list(
  Hours_only = predict(hours_fit, newdata = test),
  Time_trend = predict(time_fit,  newdata = test),
  Linear_mixed_model = predict(cv_fit, newdata = test, re.form = NULL)
)
# Compare RMSE and MAE for the same test observations.
scores <- data.frame(
  Model = names(predictions),
  RMSE = vapply(predictions, function(p) sqrt(mean((test$grade - p)^2)), numeric(1)),
  MAE  = vapply(predictions, function(p) mean(abs(test$grade - p)),       numeric(1))
)
scores
```

Population and student-specific predictions

```
# A new student with 4 study hours per day (centred: 1) in semester 4 (time = 3).
new_student <- data.frame(id = factor("new_student"), time = 3, study_hours_c = 1)
# Population prediction for a new student, without student-specific effects.
predict(final_fit, newdata = new_student, re.form = ~0, allow.new.levels = TRUE)
# Use student 1's estimated effects for a conditional prediction.
known_student <- data.frame(id = factor("1", levels = levels(long$id)),
                            time = 3,
                            study_hours_c = long$study_hours_c[long$id == "1"][1])
predict(final_fit, newdata = known_student, re.form = NULL)
```

Population predictions exclude random effects; conditional predictions include the estimated effects for a student already observed.

Generalised and nonlinear mixed models

Generalised linear mixed models (GLMM)

For repeated outcomes $Y_{i,t}$ with a distribution other than Normal, a generalised linear mixed model specifies a linear model for a function of the mean $\mu_{i,t}$:

$$Y_{i,t} \sim f(\cdot; \mu_{i,t}), \qquad g(\mu_{i,t}) = x_{i,t}^\top \beta + z_{i,t}^\top b_i, \qquad b_i \sim N(0, D).$$

$\mu_{i,t}$ is the conditional mean; $x_{i,t}$ contains the observed predictors and β their population-level effects; $z_{i,t}^\top b_i$ is the subject-specific contribution; D is the random-effects covariance matrix. Choose the distribution and link to match the outcome: Bernoulli with logit for binary outcomes, Poisson with log for counts. Outcomes are often assumed independent given b_i; integrating over b_i induces within-subject association. In R, `lme4::glmer()` fits many GLMMs.

A Bernoulli GLMM: pass/fail

At semester $t = 1, \ldots, 4$ let $Y_{i,t} = 1$ if student i earns at least 50 marks and 0 otherwise. With $T_t = t - 1$ and $H_{i,t}^c = H_{i,t} - 3$ (study hours centred at 3),

$$Y_{i,t} \mid x_{i,t}, z_{i,t}, b_i \sim \text{Bernoulli}(p_{i,t}), \qquad \text{logit}(p_{i,t}) = \beta_0 + \beta_1 T_t + \beta_2 H^c_{i,t} + b_{0,i} + b_{1,i} T_t, \qquad b_i \sim N(0, D).$$

$p_{i,t}$ is the pass probability conditional on the student's random effects; $\text{logit}(p) = \log\{p/(1-p)\}$. β_0 is the log-odds at semester 1 and 3 study hours/day when $b_{0,i} = b_{1,i} = 0$; β_1 and β_2 are conditional log-odds changes per semester and per additional study hour/day; $b_{0,i}$ and $b_{1,i}$ are student-specific departures in the baseline log-odds and the time trend, and D gives their variances and covariance.

With the linear predictor $\eta_{i,t} = \beta_0 + \beta_1 T_t + \beta_2 H^c_{i,t} + b_{0,i} + b_{1,i} T_t$, the pass probability is $\pi_{i,t} = \Pr(Y_{i,t} = 1 \mid x_{i,t}, z_{i,t}, b_i) = 1/\{1 + \exp(-\eta_{i,t})\}$. For an observed $y_{i,t} \in \{0, 1\}$ the conditional probability is

$$p(y_{i,t} \mid x_{i,t}, z_{i,t}, b_i; \beta) = \pi_{i,t}^{y_{i,t}} (1 - \pi_{i,t})^{1 - y_{i,t}} :$$

if $y_{i,t} = 1$ the contribution is $\pi_{i,t}$; if $y_{i,t} = 0$ it is $1 - \pi_{i,t}$. The four semester outcomes are conditionally independent given b_i. The likelihood for the observed data averages over the possible student-specific intercepts and slopes:

$$L(\beta, D) = \prod_{i=1}^{n} \int_{\mathcal{B}_i} \Big\{ \prod_{t=1}^{4} p(y_{i,t} \mid x_{i,t}, z_{i,t}, b_i; \beta) \Big\} f_D(b_i)\, db_i.$$

This integral has no closed form in general; `glmer()` approximates it with the Laplace method by default.

```
library(lme4)
exam <- long                                    # Complete set of student-semester records.
exam$pass <- as.integer(exam$grade >= 50)        # Code a pass as 1 when the grade is at least 50.
# Fixed effects and student-specific intercepts and time slopes.
pass_fit <- glmer(pass ~ time + study_hours_c + (time | id),
                  data = exam, family = binomial)     # Bernoulli outcomes with the logit link.
summary(pass_fit)                                # Fixed effects and random-effect variances.
```

The model relates the semester and the study hours recorded that semester to the log-odds of passing, while allowing students to differ in baseline odds and time trend.

A nonlinear mixed model: the student learning curve

Suppose grades improve quickly at first and then approach a plateau. An illustrative nonlinear mixed model is

$$Y_{it} = \frac{A + b_i}{1 + \exp\{(m - t)/s\}} + \varepsilon_{it}, \qquad b_i \sim N(0, \tau^2), \quad \varepsilon_{it} \sim N(0, \sigma^2).$$

A is the population upper grade plateau and b_i lets each student's plateau differ; m is the semester at which the curve reaches half its plateau; s controls how quickly it rises. The nonlinear time pattern represents diminishing improvement over semesters. The example uses simulated grades; the data are not real student records.

```
set.seed(2026)                                          # Reproducible simulated example.
# 800 students with grades recorded in semesters 1 to 4.
learning <- expand.grid(id = factor(1:800), semester = 1:4)
b_asym <- rnorm(800, mean = 0, sd = 5)                  # Departure from the upper grade plateau.
# Grades from a logistic learning curve plus measurement error.
learning$grade <- with(learning,
  SSlogis(semester, Asym = 90 + b_asym[as.integer(id)], xmid = 1.2, scal = 1.2)
) + rnorm(nrow(learning), mean = 0, sd = 4)

library(nlme)                                           # Provides the nlme() fitting function.
# Logistic curve with population parameters and student-specific plateaus.
learning_fit <- nlme(
  grade ~ SSlogis(semester, Asym, xmid, scal),
  data = learning,
  fixed = Asym + xmid + scal ~ 1,
  random = Asym ~ 1 | id,
  start = c(Asym = 90, xmid = 1.2, scal = 1.2)
)
```

```
summary(learning_fit)       # Population estimates and random-effect variation.

# Population-average curve on a smooth semester grid (level 0 excludes random effects).
curve_grid <- data.frame(semester = seq(1, 4, length.out = 100), id = learning$id[1])
curve_grid$fit <- predict(learning_fit, newdata = curve_grid, level = 0)
plot(learning$semester, learning$grade, pch = 16, col = rgb(0, 0, 0, alpha = 0.12),
     xlab = "Semester", ylab = "Grade (marks out of 100)")
lines(curve_grid$semester, curve_grid$fit, lwd = 2, col = "darkorange")
```

Each student contributes four repeated measurements and the generated curve levels off over time. `fixed` estimates one population plateau, midpoint and scale; `random` lets the plateau vary across students. The fitted curve summarises the population pattern; student-specific curves would also include the estimated random effects.

Other correlation structures, workflow and take-aways

Other correlation structures for repeated outcomes

- **Independence:** no working correlation between repeated outcomes.
- **Exchangeable:** the same correlation for every pair of occasions.
- **AR(1):** correlation decreases as occasions become further apart.
- **Unstructured:** each pair has its own correlation; this requires enough data.

The working structure should reflect the timing of measurements and the study design (these structures are developed in Part III).

A practical workflow for longitudinal analysis

1. **Define the problem:** specify who is followed, the outcome, the time scale and the target (association, change, or prediction).
2. **Choose a model:** inspect the scatterplot and the student trajectories; choose the fixed effects and a plausible random-effects structure.
3. **Fit the model:** use `lmer()`, and ML when comparing nested fixed-effects structures.
4. **Check fit:** inspect residuals, convergence, singularity, and prediction on held-out students.
5. **Interpret heterogeneity:** examine the random-intercept and random-slope estimates and their uncertainty.
6. **Predict:** distinguish a population prediction for a new student from a conditional prediction for an observed student.

At every stage, state the assumptions explicitly.

Take-away messages

- Longitudinal data contain repeated measurements on the same people; those records are usually dependent.
- Fixed effects describe the average trajectory and covariate associations.
- Random effects describe student-specific departures and induce within-student correlation.
- `lme4` supports model fitting, comparison, diagnostics, and population or student-specific prediction.
- Conclusions depend on the time scale, the outcome model, the diagnostics and the prediction target.

Further reading (Part II)

- Fitzmaurice, Laird and Ware. *Applied Longitudinal Analysis*; Harvard lecture slides.
- Diggle, P. *Longitudinal Data Analysis*; Belfast lecture notes.

- Washington SISCER, GEE and mixed-effects models.
- Guindani, M. *Statistical Methods for Correlated Data*, lecture slides.
- Bates et al. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48.
- Official `lmer()` and prediction documentation.
- Ho Thi Nhan, *Longitudinal Data Analysis with R*, Chapter 9.

Part III

Complementary Notes: Correlated Data and Linear Mixed Models

Lecture 2 material · "Introduction to linear mixed effects models" · M. Guindani, UCLA

Preface and course context

These slides are part of a 10-week course taught at UCI in Spring 2022. They provide an introduction to statistical methods for analysing correlated data; topics include linear mixed models, nonlinear mixed-effects models and generalised estimating equations. The content is borrowed mostly from books on the topics. The main reference (recommended text) was J. Wakefield (2013), *Bayesian and Frequentist Regression Methods*, Chapter 8 and following, Springer. Other books heavily used:

1. Fitzmaurice, Davidian, Verbeke and Molenberghs (2009). *Handbook of Longitudinal Data Analysis*, CRC Press.
2. Fitzmaurice, Laird and Ware (2004). *Applied Longitudinal Data Analysis*, 2nd ed., Wiley.
3. Diggle, Heagerty, Liang and Zeger (2002). *Analysis of Longitudinal Data*, Oxford University Press.
4. McCulloch and Neuhaus (2001). *Generalized Linear Mixed Models*, Wiley.
5. Song (2007). *Correlated Data Analysis*, Springer.
6. Wood (2017). *Generalized Additive Models: An Introduction with R*, 2nd ed., CRC Press.
7. Weiss (2005). *Modeling Longitudinal Data*, Springer Texts in Statistics.

The author notes that typos and errors are his own, and invites readers to point out any missing credit.

Introduction: correlated data

With linear and generalised linear models, the models assume independence (conditional independence) between observations:

$$Y_1, \dots, Y_n \mid X, \beta, \sigma^2 \overset{\text{ind}}{\sim} N(X_i\beta, \sigma^2),$$

where $X_i\beta = \mu_i$ is the regression line capturing the relationship between the pairs (x_i, y_i), $i = 1, \dots, n$. Here we consider models for *dependent* data arising from (1) sampling over time (e.g. blood pressure over several days), (2) sampling in space (e.g. diseased subjects in different counties; split-plot designs), and (3) sampling within clusters (e.g. families).

Longitudinal, time-series and clustered data

- **Longitudinal data analysis** estimates how individuals' measurements change over time (e.g. over the duration of a study) and examines the factors that influence heterogeneity among individuals ("what spurs" the changes over time). The goal is to characterise change in the response over time as a function of a subset of covariates (repeated-measures analysis).
- **Time-series data** are typically high-frequency (high-resolution) data, studied with specialised techniques (omitted here).
- **Clustered data:** observations on the same unit exhibit residual dependence due to shared unmeasured variables, after controlling for known regressors.

Why specific analytic tools? Two examples

The MIT Growth and Development Study

Body fatness in girls is thought to increase just before or around menarche (first menstrual period), levelling off approximately 4 years after menarche. Researchers want to determine the increase in body fatness after menarche (*how might you design a study to investigate this question?*). The MIT Growth and Development Study is a prospective study on body-fat accretion in a cohort of 162 girls:

- at the start of the study all girls were pre-menarcheal and non-obese (triceps skinfold thickness below the 85th percentile);
- all girls were followed according to a schedule of annual measurements until four years after menarche;
- at each examination a measure of body fatness was obtained from bioelectric impedance analysis and the percent body fat (PBF) was derived;
- in total 1,049 individual PBF measurements, an average of 6.4 per subject.

The plot of PBF (from 0 to about 50) against time relative to menarche (from -6 to $+4$ years) shows body fat rising around menarche.

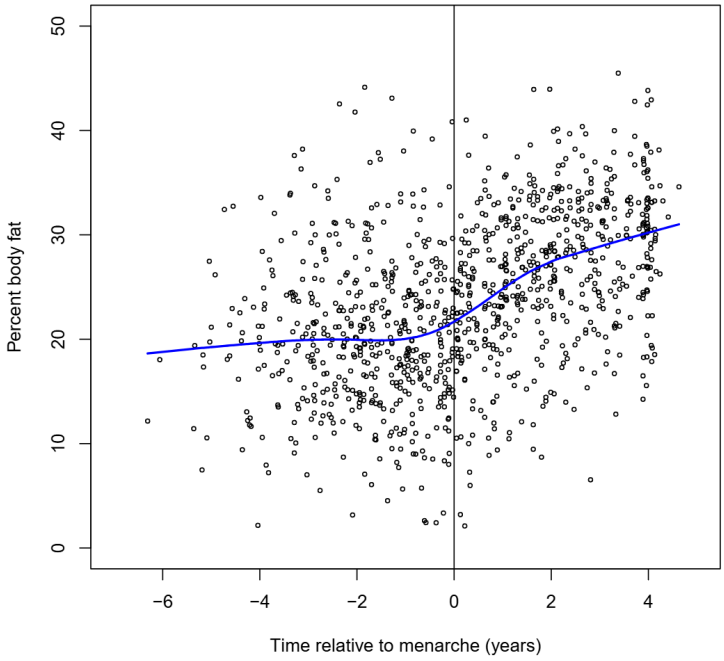


Figure 5. MIT Growth and Development Study: percent body fat (PBF) against time relative to menarche (years), 1,049 measurements on 162 girls. The vertical line marks menarche; the blue curve is a smooth fit.

A naïve solution

```
summary(lm(formula=PBF~Time.M))

Residuals:
    Min      1Q  Median      3Q     Max
-21.5499  -4.5766   0.2428   4.7401  23.5149

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 23.33576    0.22157  105.32   <2e-16 ***
Time.M       1.47321    0.09459   15.57   <2e-16 ***

Residual standard error: 7.153 on 1047 degrees of freedom
Multiple R-squared: 0.1881,  Adjusted R-squared: 0.1873
F-statistic: 242.6 on 1 and 1047 DF,  p-value: < 2.2e-16
```

What is wrong with this analysis? It treats the 1,049 measurements as independent, although they come from only 162 girls; the line plot that joins each girl's measurements makes the within-girl dependence visible.

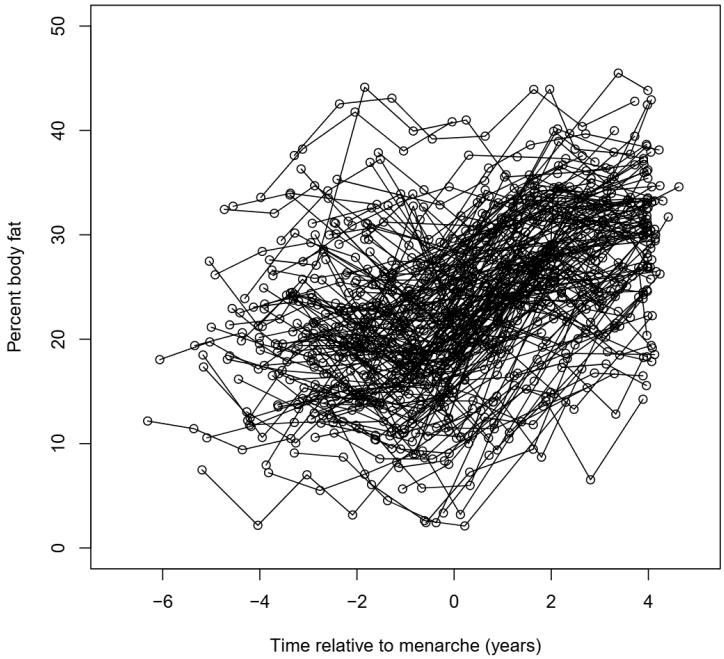


Figure 6. The same data with the measurements of each girl joined by line segments (line plot): repeated measurements on the same girl are visibly related.

A better comparison: the paired t-test

Look at each girl's first (pre-menarche) and last (post-menarche) PBF measurements and test whether the mean gain is positive.

```
t.test(PostPBF, PrePBF, paired=TRUE, alternative="greater")
```

```
Paired t-test
data:  PostPBF and PrePBF
t = 17.521, df = 159, p-value < 2.2e-16
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 7.195989      Inf
sample estimates:
mean difference
       7.946375
```

Assumptions. Normality of the differences? The data show heavy-tailed distributions (normal QQ plot), but the test is robust for large samples, and here $N = 162$. Independence? Fine, since each difference was measured on a different girl.

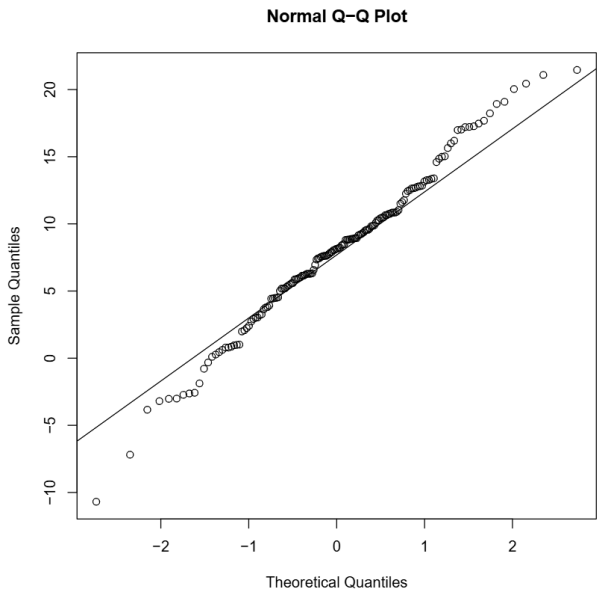


Figure 7. Normal Q–Q plot of the post-menarche minus pre-menarche PBF differences: the tails are heavier than normal, but the paired t-test is robust for large samples.

Limitations. Not all girls were measured at the same time relative to menarche: pre-menarche measurements range from 0.13 to 6.31 years before menarche, post-menarche measurements from 0.38 to 4.63 years after; two girls had their last measurements at 0.03 and 0.04 years *before* menarche and those two data points had to be discarded. The test only checks whether there is a significant mean difference between first and last measurements; it cannot model how PBF changes over time. We would like to incorporate the positive correlation between repeated measurements on the same individual into the model.

Dental growth curves

Dental measurements of the distance (mm) from the centre of the pituitary gland to the pterygo-maxillary fissure are available for 11 girls and 16 boys at ages 8, 10, 12 and 14. For now we concentrate on the girls. The data are the `Orthodont` dataset of the `nlme` package (grouped data `distance ~ age | Subject`, with columns `distance, age, Subject, Sex`).

```
library(nlme)
data(Orthodont)
head(Orthodont, n=15)
```

Table 13. First rows of `Orthodont` (subjects M01–M04 shown; the girls are the subjects labelled F).

distance	age	Subject	Sex
26.0	8	M01	Male
25.0	10	M01	Male
29.0	12	M01	Male
31.0	14	M01	Male
21.5	8	M02	Male
22.5	10	M02	Male
23.0	12	M02	Male
26.5	14	M02	Male
23.0	8	M03	Male
22.5	10	M03	Male
24.0	12	M03	Male
27.5	14	M03	Male
25.5	8	M04	Male
27.5	10	M04	Male
26.5	12	M04	Male

The "spaghetti plot" (Figure 8.1 of Wakefield's textbook) shows the profiles of the distance measurements of the 11 girls over the years: the slopes look quite similar, though there is clearly between-girl variability in the intercepts.

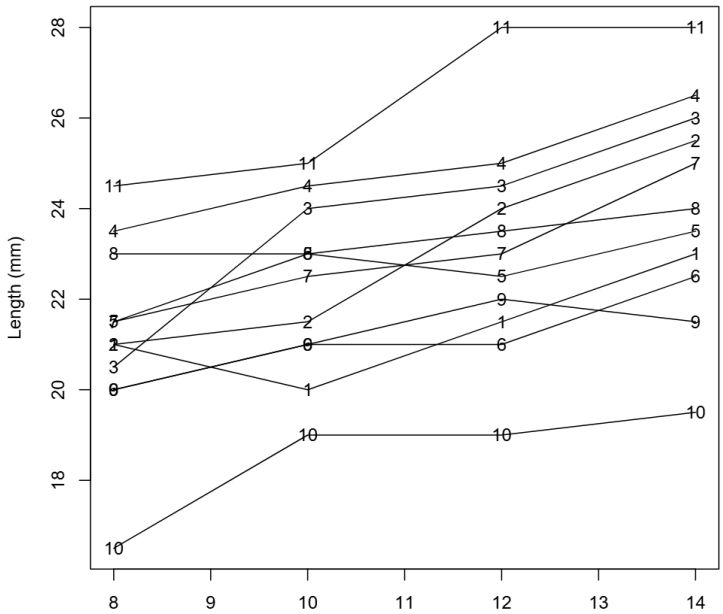


Figure 8. Dental growth curves (`Orthodont`, girls): distance (mm) at ages 8, 10, 12 and 14 years for the 11 girls; the numbers label the girls (age in years on the horizontal axis). The slopes look similar; the intercepts vary between girls.

Potential objectives of the analysis:

1. *Population inference:* describe the average growth as a function of age for the population from which the sample of children was selected.
2. *Within- versus between-child variability:* assess the within- to between-child variability in growth measurements.
3. *Individual-level inference:* for a child in the sample, or for a new unobserved child from the same population (a "growth chart").

Linear mixed models and marginal models

We discuss mixed-effects models, which contain both *fixed effects* shared by all individuals and *random effects* unique to particular individuals and assumed to arise from a distribution. The linear mixed-effects model allows the estimation of a single curve for each girl. By marginalising over the random effects we obtain marginal (population-wide) inference. Let $\boldsymbol{Y}_{ij}$ be the $\boldsymbol{j}$th measurement, taken at time $\boldsymbol{t}_j$, on the $\boldsymbol{i}$th child, $\boldsymbol{i = 1, \ldots, m = 11}$, $\boldsymbol{j = 1, \ldots, n_i = 4}$, and consider the marginal model

$$\mathbb{E}[\boldsymbol{Y}_{ij}] = \beta_0^M + \beta_1^M \boldsymbol{t}_j,$$

with β_0^M, β_1^M the marginal intercept and slope. The *marginal residuals* $\boldsymbol{e}_{ij}^M = \boldsymbol{Y}_{ij} - \beta_0^M - \beta_1^M \boldsymbol{t}_j$ are not expected to be independent, because of the dependence of observations on the same girl. Let the matrix with the standard deviations $\sigma_j = \sqrt{\mathbf{var}(e_{ij}^M)}$ on the diagonal and the correlations

$$\rho_{jk} = \frac{\mathbf{cov}(e_{ij}^M, e_{ik}^M)}{\sqrt{\mathbf{var}(e_{ij}^M)\mathbf{var}(e_{ik}^M)}}, \qquad j \neq k,$$

below it represent the residuals' standard deviations and correlations (assumed constant across girls). Fitting the marginal model to these data and estimating the entries empirically gives

$$\begin{pmatrix} 2.12 & & & \\ 0.83 & 1.90 & & \\ 0.86 & 0.90 & 2.36 & \\ 0.84 & 0.88 & 0.95 & 2.44 \end{pmatrix} \tag{1}$$

showing a clear correlation between residuals at different ages on the same girl. Hence methods for independent data that assume zero within-girl correlation will give inappropriate standard errors and uncertainty estimates for $\hat{\beta}_0^M$ and $\hat{\beta}_1^M$.

Take-away points

- Fitting marginal models allows population-wide inference: it allows the direct assessment of the average responses at different times.
- Marginal models require minimal assumptions: only the mean function (and the correlations) were used.
- Marginal models require reliable estimation of the correlation functions in order to model the within-individual correlations.

As a counterpoint, linear mixed models estimate a curve for each child while "borrowing strength" in the estimation across children. One could consider a separate estimate for each curve (all fixed effects, no random part); the linear mixed model instead estimates population-level parameters (fixed effects) while accommodating individual variation. In the frequentist setting random effects are typically seen as a convenient tool to model within- and between-individual variability, often with no interest in the individual random effects' values. In the Bayesian framework random effects are latent variables with a

prior, also used to model unobserved correlation, and there is often real interest in their posterior estimates.

Correlated data and longitudinal studies

What are correlated data?

We look at "clustered" data: observations within each cluster are correlated with each other. *Positive correlation*: large measurements tend to cluster with large measurements. *Negative correlation*: large measurements tend to cluster with small ones. Examples of clusters: repeated measurements on the same individual over time (longitudinal data); measurements on a mother and a daughter; measurements on all individuals in a household.

Cross-sectional versus longitudinal

- In a **cross-sectional** study, measurements are obtained at a single point in time; it is not possible to assess individual changes across time.
- In a **longitudinal** study, participants are measured repeatedly throughout the study, which permits direct assessment of changes in the response over time. Participants or units are individuals or subjects, measured at different times or occasions that need not be equally spaced. The defining feature is that two or more observations of the response, at different times, are made on at least some participants.

Terminology

- If all individuals have the same number of repeated measurements at a common set of occasions, the study is **balanced** over time. Otherwise (different occasions or numbers of measurements) it is **unbalanced**; this is common when a study is retrospective (e.g. data from medical databases) or when times are defined relative to an individual benchmark event, as in the menarche study.
- If there are missing data (an intended measurement could not be obtained), the data set is called **incomplete**. Missing data are the rule, not the exception, in longitudinal studies in the health sciences: participants do not always appear for a scheduled observation or leave the study before completion. With missing data the dataset is necessarily unbalanced (see Part I).

Goals of longitudinal studies

There are two goals: (i) assess *within-individual* (intra-individual) changes of the response (how to characterise the change over time); (ii) assess *between-individual* (inter-individual) differences (are the response trajectories related to certain covariates?). Cross-sectional studies can only assess between-individual variation.

Correlation in longitudinal data

The correlations among repeated measures on one individual are (1) positive; (2) decreasing with increasing time separation; (3) rarely approaching zero for pairs far apart in time; (4) rarely approaching one for pairs very close in time.

Sources of variation

1. **Between-subject variability.** *Heterogeneity in the mean response:* some individuals consistently respond higher than average and others lower (annual income, daily caloric intake, systolic blood pressure); this induces a positive correlation between repeated measurements. *Heterogeneity in the response trajectory:* some individuals improve faster than others and some may worsen (CD4 counts after antiviral treatment, rate of increase in income); this often induces decreasing correlations with

increasing time separation, e.g. scores at times 1 and 4 are often less correlated than at times 1 and 2. In statistical models between-individual variability is accounted for by individual-specific random effects (randomly varying intercepts and slopes).

2. **Within-subject variability.** Repeated measures are realisations of a biological process operating within the individual (weight, systolic blood pressure, serum cholesterol). *Serial correlation*: a stronger correlation for measurements closer in time, because the underlying process changes through time in a relatively smooth and continuous fashion.
3. **Measurement error.** Not to be confused with within-subject biological variation; it may shrink the correlation among repeated measures closer to zero.

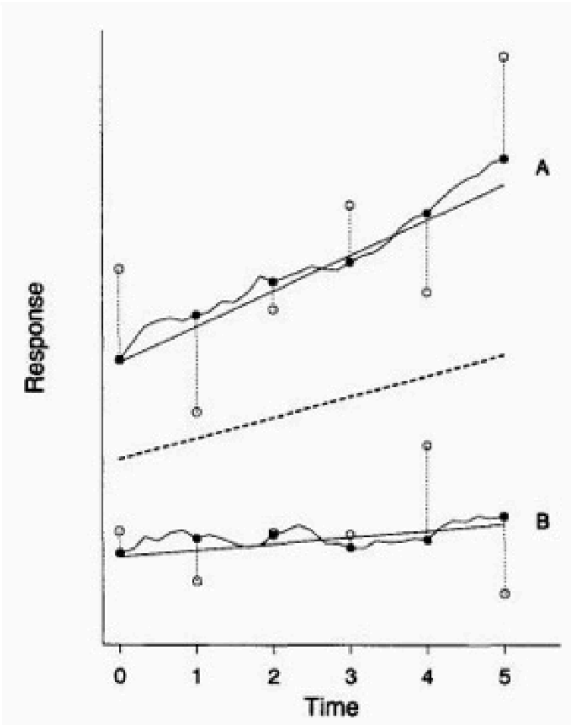


Figure 9. Graphical representation of the three sources of variability in longitudinal data for two hypothetical individuals, A and B: (1) between-individual heterogeneity, (2) within-individual biological variation, (3) measurement error. Filled dots: repeated measures free of measurement error; open circles: observed repeated measures with measurement error. Straight solid line: true individual response trajectory (free of biological variation); jagged curve: within-individual biological variation around it; dashed line: average true response trajectory of the two respondents.

What if we ignored the nature of the data? Cross-sectional versus longitudinal effects

Consider a response that changes over time and varies among subjects (age, blood pressure, weight), measured on n occasions (j) on m individuals (i), with the simple model

$$Y_{ij} = \beta_0 + \beta x_{ij} + \varepsilon_{ij}, \qquad j = 1, \ldots, n;\ i = 1, \ldots, m. \tag{2}$$

It can be re-expressed as $Y_{ij} = \beta_0 + \beta x_{i1} + \beta(x_{ij} - x_{i1}) + \varepsilon_{ij}$, which makes explicit the assumption that the *cross-sectional* effect due to x_{i1} equals the *longitudinal* effect represented by $x_{ij} - x_{i1}$; this is a strong assumption, doomed to fail in many studies. Giving each person their own intercept, $Y_{ij} = \beta_{0i} + \beta(x_{ij} - x_{i1}) + \varepsilon_{ij}$, is the opposite extreme (each baseline different). An intermediate and more useful model is

$$Y_{ij} = \beta_0 + \beta^C x_{i1} + \beta^L (x_{ij} - x_{i1}) + \varepsilon_{ij}. \tag{3}$$

The term $\beta^C x_{i1}$ allows cross-sectional and longitudinal effects to be examined separately, and one can test whether they are the same ($\beta^C = \beta^L$). x_{i1} can be seen as a confounding variable whose absence may bias the estimate of the true longitudinal effect.

Bias. Under model (2) the least-squares estimate is

$$\hat{\beta} = \sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-\bar{x})(y_{ij}-\bar{y}) \Big/ \sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-\bar{x})^2,$$

with $\bar{x} = \sum_{ij} x_{ij}/(nm)$ and $\bar{y} = \sum_{ij} y_{ij}/(nm)$. If the true model is (3), then

$$\mathbb{E}(\hat{\beta}) = \beta^L + \frac{\sum_{i=1}^{m} n\,(x_{i1}-\bar{x}_1)(\bar{x}_i-\bar{x})}{\sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-\bar{x})^2}\,(\beta^C-\beta^L),$$

where $\bar{x}_i = \sum_j x_{ij}/n$ and $\bar{x}_1 = \sum_i x_{i1}/m$. Hence the cross-sectional estimate $\hat{\beta}$, which assumes $\beta^L = \beta^C$, is a biased estimate of β^L, unbiased only if $\beta^L = \beta^C$ or the variables $\{x_{i1}\}$ and $\{\bar{x}_i\}$ are orthogonal. The direction of the bias depends on the correlation between x_{i1} and $\bar{x}_i$.

The efficiency of longitudinal designs

Designs that collect dependent data can be very efficient: applying different treatments to the same patient over time can be very beneficial, since each patient acts as their own control. Suppose we compare two treatments, coded -1 and $+1$, with four measurements in total.

Cross-sectional study. One measurement on each of four individuals, two (i = 1, 2) on treatment -1 and two (i = 3, 4) on treatment $+1$: $Y_{i1} = \beta_0 + \beta_1 x_{i1} + \varepsilon_{i1}$, $i = 1,\dots,4$. The treatment effect is $\mathbb{E}[Y_1 \mid x=1] - \mathbb{E}[Y_1 \mid x=-1] = 2\beta_1$, with unbiased OLS estimators

$$\hat{\beta}_0^c = \frac{\sum_{i=1}^{4} Y_{i1}}{4}, \qquad \hat{\beta}_1^c = \frac{Y_{31}+Y_{41}-(Y_{11}+Y_{21})}{4}, \qquad \operatorname{var}(\hat{\beta}_1^c) = \frac{\sigma^2}{4}.$$

Longitudinal study. Two observations on each of two individuals (again four measurements): $Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_i + \delta_{ij}$, with b_i (individual effect) and δ_{ij} (measurement error) independent, $\mathbb{E}[\delta_{ij}] = 0$, $\operatorname{var}(\delta_{ij}) = \sigma_\delta^2$, $\mathbb{E}[b_i] = 0$, $\operatorname{var}(b_i) = \sigma_0^2$. Marginalising over b_i,

$$\mathbb{E}(Y_{ij}) = \beta_0 + \beta_1 x_{ij}, \quad \operatorname{Var}(Y_{ij}) = \sigma_0^2 + \sigma_\delta^2 = \sigma^2, \quad \operatorname{Cov}(Y_{ij}, Y_{ik}) = \operatorname{Var}(b_i) = \sigma_0^2.$$

With $Y = [Y_{11}, Y_{12}, Y_{21}, Y_{22}]$, $\operatorname{var}(Y) = \sigma^2 R$ with

$$R = \begin{pmatrix} 1 & \rho & 0 & 0 \\ \rho & 1 & 0 & 0 \\ 0 & 0 & 1 & \rho \\ 0 & 0 & \rho & 1 \end{pmatrix}, \qquad \rho = \sigma_0^2/\sigma^2$$

the correlation between observations on the same individual. By generalised least squares, $\hat{\beta}^L = (x^\top R^{-1} x)^{-1} x^\top R^{-1} Y$ is unbiased, with $\operatorname{var}(\hat{\beta}^L) = (x^\top R^{-1} x)^{-1}\sigma^2$. It is easy to show that the relative efficiency of the longitudinal design is

$$\frac{\operatorname{var}(\hat{\beta}_1^L)}{\operatorname{var}(\hat{\beta}_1^c)} = \frac{1-\rho^2}{1-\rho\,(x_{11}x_{12}+x_{21}x_{22})/2}.$$

- **Constant treatment within each individual** ($x_{11} = x_{12} = -1$, $x_{21} = x_{22} = 1$): the ratio is $1+\rho$, so the cross-sectional study is preferable in the usual situation in which observations on the same individual are positively correlated (there is a benefit from adding more subjects).

- **Varying treatment within each individual** ($x_{11} = x_{22} = 1$, $x_{12} = x_{21} = -1$): the ratio is $1 - \rho$, so the longitudinal study is more efficient when $\rho > 0$, because each individual acts as their own control.

These results extend to time-varying covariates.

Linear mixed models

Basic idea. The measurements on each unit (individual) are explained by a regression model combining *fixed effects*, common to all units of the population, and *random effects*, unit-specific perturbations. Let $Y_i = [Y_{i1}, \dots, Y_{in_i}]^\top$, $i = 1, \dots, m$, be the measurements over n_i occasions on subject i; $x_{ij} = [1, x_{ij1}, \dots, x_{ijk}]$ the $(k+1)$ vector of covariates at occasion j and $x_i = [x_{i1}, \dots, x_{in_i}]$ the design matrix of unit i; $z_{ij} = [1, z_{ij1}, \dots, z_{ijq}]^\top$ a $(q+1)$ vector of variables (e.g. a subset of x_{ij}) and $z_i = [z_{i1}, \dots, z_{in_i}]^\top$ the design matrix.

A linear mixed model is described in two levels (stages):

1. **Conditional model for the response:** $y_i = x_i\beta + z_i b_i + \varepsilon_i$, where ε_i is an $n_i \times 1$ zero-mean vector of error terms, $i = 1, \dots, m$.
2. **Assumptions on the random components:** $\mathbb{E}[\varepsilon_i] = 0$, $\mathbf{var}(\varepsilon_i) = E_i(\alpha)$; $\mathbb{E}[b_i] = 0$, $\mathbf{var}(b_i) = D(\alpha)$; $\mathbf{cov}(b_i, \varepsilon_{i'}) = 0$ for all i, i', where α is an $r \times 1$ vector collecting the variance–covariance parameters; further $\mathbf{cov}(\varepsilon_i, \varepsilon_{i'}) = 0$ and $\mathbf{cov}(b_i, b_{i'}) = 0$ for $i \neq i'$.

Averaging over the random effects,

$$\mathbb{E}[Y_i] = x_i\beta, \qquad \mathbf{var}(Y_i) = V_i(\alpha) = z_i D(\alpha) z_i^\top + E_i(\alpha),$$

for $i = 1, \dots, m$, where $V_i(\alpha)$ is $n_i \times n_i$. The random effects induce a dependence structure across within-individual measurements: because b_i is fixed by group, not varying with the observation, within-group observations share the same random effects and are therefore correlated. The diagonal elements of $z_i D z_i'$ need not be constant, so the set-up also accommodates heteroscedasticity. Still, $\mathbf{cov}(Y_i, Y_{i'}) = 0$ for $i \neq i'$.

The within-group error contribution to the marginal covariance is given directly by $E_i(\alpha)$, which can be non-diagonal (correlation) with different diagonal elements (heteroscedasticity). The matrices $E_i(\alpha)$ are typically assumed to depend on i only through their dimensions, being parameterised by a fixed, generally small, set of parameters (e.g. $\sigma^2 I_n$ for independent errors, or an AR(1) structure; Box et al., 1994). Even though the random effects are useful and intuitive quantities, they are not observable; likelihood estimation and inference in the frequentist setting therefore rely on the marginal distribution of the observed response vectors y_i.

Example: ANOVA models

One-way random-effects model. A model is a (pure) random-effects model if the only fixed effect is an unknown mean:

$$y_{ij} = \mu + b_i + \varepsilon_{ij}, \qquad b_i \overset{\text{iid}}{\sim} N(0, \sigma_0^2),\ \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

that is $b \sim N(0, \sigma_0^2 I_n)$ and $\varepsilon \sim N(0, \sigma^2 I_n)$, with $\alpha = (\sigma_0^2, \sigma^2)$. **Two-way random-effects model:** $y_{ij} = \mu + \xi_i + \eta_j + \varepsilon_{ij}$ with $\xi_i \sim N(0, \sigma_1^2)$, $\eta_j \sim N(0, \sigma_2^2)$, and independent errors $\varepsilon_{ij} \sim N(0, \tau^2)$; again the random effects and errors are independent.

General ANOVA model. $y_{ij} = \mu + Zb_i + \varepsilon_{ij}$ with $Zb_i = Z_{1i}b_{1i} + \cdots + Z_{is}b_{is}$, where $b_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_j^2)$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \tau^2)$, and the b_i and ε_{ij} are independent. A natural set of variance components is $\tau^2, \sigma_1^2, \dots, \sigma_s^2$ (the *original form*); alternatively the *Hartley–Rao form* (1967) is $\lambda = \tau^2$, $\gamma_1 =$

$\sigma_1^2/\tau^2, \ldots, \gamma_s = \sigma_s^2/\tau^2$. Marginalising the random vectors, $y \sim N(\mu, V)$ with $V = \tau^2 I_n + \sum_{i=1}^{s} \sigma_i^2 Z_i Z_i'$, which again characterises the dependence induced by the random effects.

The random-intercept model

Another very simple and much used model in longitudinal data analysis, in which the observations may depend on covariates:

$$y_{ij} = x_{ij}'\beta + b_i + \varepsilon_{ij}, \qquad b_i \sim N(0, \sigma_b^2),\ \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2).$$

b_i describes how the mean response of individual i deviates from the population average; ε_{ij} is the measurement or sampling error. This is the model behind the so-called "repeated-measures ANOVA":

$$\mathrm{Var}(y_{ij}) = \sigma_b^2 + \sigma_\varepsilon^2, \qquad \mathrm{Cov}(y_{ij}, y_{ik}) = \sigma_b^2,$$

a covariance structure called *compound symmetry* (see below).

The random-intercept and slope model

If the mean response changes only as a function of time (no other covariates),

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{0i} + b_{1i} t_{ij} + \varepsilon_{ij}, \qquad j = 1, \ldots, n_i,\ i = 1, \ldots, n.$$

Each subject's mean response varies not only in the baseline response ($t_{i1} = 0$) but also in the change of the response over time. Individuals may have a lower or higher baseline ($\beta_0 + b_{0i}$) than the population average (β_0), and a steeper or less steep rate of change ($\beta_1 + b_{1i}$) than the population average (β_1). In the lecture figure the population mean is drawn in black and the individual trajectories around it.

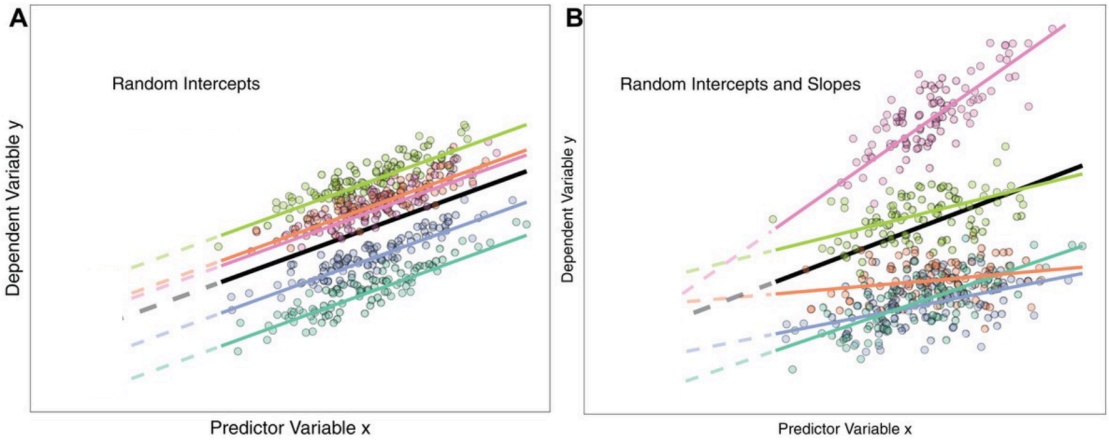


Figure 10. (A) Random-intercepts model and (B) random-intercepts-and-slopes model. Coloured lines are individual trajectories and points their observations; the black line is the population mean. Dashed segments extrapolate the lines back towards the origin.

Interpretation of the parameters. The baseline need not be $t = 0$ in the data; more generally it can be a specified time $\bar{t}$. For a generic individual at time t the conditional model is $\mathbb{E}[Y \mid b, t] = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})(t - \bar{t})$ and the marginal model is $\mathbb{E}[Y \mid t] = \beta_0 + \beta_1(t - \bar{t})$, so that β_0 is the expected (population-averaged) response at $t = \bar{t}$ and β_1 the expected (population-averaged) change per unit increase in time. Alternatively β_1 is the change per unit time for a "typical" individual ($b_1 = 0$). For a specific individual i, $\beta_0 + b_{0i}$ is the expected response at $t = \bar{t}$ and $\beta_1 + b_{1i}$ the expected change per unit increase in time.

Remark 7 (covariates in intercepts and slopes). The effects of covariates (treatments, exposures, background characteristics) can be incorporated by letting the means of the intercepts and slopes depend on them. For a two-group study (treatment and control) with an approximately linear mean response over time, whose intercepts and slopes depend on group:

$$Y_{ij} = \beta_1 + \beta_2 t_{ij} + \beta_3 \mathrm{Group}_i + \beta_4 t_{ij} \times \mathrm{Group}_i + b_{1i} + b_{2i} t_{ij} + \varepsilon_{ij},$$

where $\mathrm{Group}_i = 1$ if the ith individual was assigned to treatment and 0 otherwise. The design matrix X_i has rows $(1, t_{ij}, 0, 0)$ in the control group and $(1, t_{ij}, 1, t_{ij})$ in the treatment group, $j = 1, \ldots, n_i$, while Z_i has rows $(1, t_{ij})$ in both groups.

Remark 8 (covariance of the random intercept-and-slope model). With an unstructured covariance matrix for the random effects, $\mathrm{Var}(b_{1i}) = g_{11}$, $\mathrm{Var}(b_{2i}) = g_{22}$, $\mathrm{Cov}(b_{1i}, b_{2i}) = g_{12}$:

$$\mathrm{Var}(Y_{ij}) = \mathrm{Var}(b_{1i} + b_{2i} t_{ij} + \varepsilon_{ij}) = g_{11} + 2 t_{ij} g_{12} + t_{ij}^2 g_{22} + \sigma^2,$$

$$\mathrm{Cov}(Y_{ij}, Y_{ik}) = g_{11} + (t_{ij} + t_{ik}) g_{12} + t_{ij} t_{ik} g_{22}.$$

Thus $\mathrm{Cov}(Y_i)$ is a function of time t_{ij}: with random intercepts and slopes the variance can increase or decrease over time as a quadratic function of the measurement times. It increases over time (for $t_{ij} \geq 0$) when $\mathrm{Cov}(b_{1i}, b_{2i}) \geq 0$ but can decrease when $\mathrm{Cov}(b_{1i}, b_{2i}) < 0$. Similarly the magnitude of the covariance (and correlation) between Y_{ij} and Y_{ik} depends on the time separation between them.

Baseline and centring

The covariance matrix of the random effects is not invariant to a linear transformation of z_i: such transformations alter the interpretation of b_i and change the estimated variances and covariances of the random effects. In the random intercept-and-slope model, centring the times ($t_{ij} - \bar{t}$, $\bar{t} \neq 0$) changes the interpretation of the intercepts, hence the estimated variance of the random intercepts and their covariance with the random slopes: with untransformed times the variance of the intercepts measures the between-subject variability at time zero; with times centred at $\bar{t}$ it measures it at time $\bar{t}$.

Linear transformations of components of z_i produce equivalent mixed-effects models only when D is unstructured. In addition, centring should be done with care: t_{ij} as time from baseline (enrolment) is fine; with t_{ij} as age be careful, since centring at an individual's age at the jth occasion (e.g. their mean over occasions) does not make sense. *Rule of thumb:* centre the times of all subjects at some common fixed age within the age range of the participants ($t_{ij} - a$, e.g. the time of menarche). By centring at a common value, the intercept is interpretable as the mean response at that common value of time (or age) and $\mathrm{Var}(b_{1i})$ also has a meaningful interpretation.

Covariance models for clustered data

Consider again $y_i = x_i \beta + z_i b_i + \varepsilon_i$ with $b_i \overset{\text{iid}}{\sim} N_{q+1}(0, D)$ and $\varepsilon_i \overset{\text{ind}}{\sim} N_{n_i}(0, E_i)$, and the common simplification of uncorrelated errors, $E_i = \sigma^2 I_{n_i}$ (σ^2 is the *measurement-error variance*). The marginal variances and covariances can be assessed directly from the observed data.

Exchangeable (compound symmetry)

For the random-intercepts-only model $z_i b_i = 1_{n_i} b_i$ with $\mathrm{var}(b_i) = \sigma_0^2$ and $E_i = \sigma_\varepsilon^2 I_{n_i}$,

$$\mathrm{var}(Y_i) = \sigma^2 \begin{pmatrix} 1 & \rho & \rho & \cdots & \rho \\ \rho & 1 & \rho & \cdots & \rho \\ \rho & \rho & 1 & \cdots & \rho \\ \vdots & & & \ddots & \vdots \\ \rho & \rho & \rho & \cdots & 1 \end{pmatrix}, \qquad \sigma^2 = \sigma^2_\varepsilon + \sigma^2_0,\ \rho = \sigma^2_0/\sigma^2.$$

The whole covariance function is reduced to two variance parameters, $\alpha = [\sigma^2_\varepsilon, \sigma^2_0]$. It is therefore not very flexible: it postulates a constant correlation across occasions, whereas correlation is typically expected to decrease. It is particularly appropriate for clustered data with no time ordering (a split-plot design, or multiple measurements within a family); for longitudinal data only over short time scales.

Serial correlation

An extension of the two-stage model is $y_i = x_i\beta + z_i b_i + \delta_i + \varepsilon_i$, with b_i the individual-specific random effects, δ_i the serial dependence and ε_i the measurement error, where $\mathbb{E}[\varepsilon_i] = 0$, $\mathrm{var}(\varepsilon_i) = \sigma^2_\varepsilon I_{n_i}$; $\mathbb{E}[b_i] = 0$, $\mathrm{var}(b_i) = D$; $\mathbb{E}[\delta_i] = 0$, $\mathrm{var}(\delta_i) = \sigma^2_\delta R_i$; and b_i, δ_i, ε_i are uncorrelated with each other, and with those of other individuals.

Autoregressive structure. A wide variety of time-varying correlations is obtained by modelling R_i. For an AR(1) process $\delta_{ij} = \rho\delta_{i,j-1} + u_{ij}$, $\mathbb{E}[u_i] = 0$, $\mathrm{var}(u_i) = \sigma^2_u I_{n_i}$, and equally spaced times,

$$R_i = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n_i-1} \\ \rho & 1 & \rho & \cdots & \rho^{n_i-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{n_i-3} \\ \vdots & & & \ddots & \vdots \\ \rho^{n_i-1} & \rho^{n_i-2} & \rho^{n_i-3} & \cdots & 1 \end{pmatrix}.$$

Toeplitz. The more general structure $\mathrm{var}(\delta_i) = \sigma^2_\delta\,[\rho_{|j-k|}]$ with $\rho_0 = 1$ and free $\rho_1, \ldots, \rho_{n_i-1}$ is most appropriate when measurements are made at equal intervals and does not take the distances between measurements into account.

Exponential covariance function. The model extends to unequally spaced times with $\mathrm{cov}(\delta_{ij}, \delta_{ik}) = \sigma^2_\delta \rho^{|t_{ij} - t_{ik}|}$: correlations decline as the separation between pairs of repeated measures increases, although the decay is quite fast with respect to what is seen in longitudinal data.

Unstructured. Allows different variances at each occasion $\sigma^2_{\delta 1}, \ldots, \sigma^2_{\delta n_i}$ and a distinct correlation for each pair, $\rho_{jk} = \rho_{kj}$, i.e. $n_i(n_i+1)/2$ parameters per individual: a large number if n_i is large. With a common design across individuals it may be plausible to fit this model, but a large number of individuals m is still needed for reliable inference. As usual, there is a trade-off between flexibility and parsimony.

Table 14. The three most common covariance structures (example for five occasions; the slide gives the upper triangles of these symmetric matrices, with ρ unknown and $|\rho| < 1$).

Structure	Example	Parameters
Compound symmetry (CS)	$\sigma^2 \begin{pmatrix} 1 & \rho & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho & \rho \\ \rho & \rho & 1 & \rho & \rho \\ \rho & \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & \rho & 1 \end{pmatrix}$	2
First-order autoregressive [AR(1)]	$\sigma^2 \begin{pmatrix} 1 & \rho & \rho^2 & \rho^3 & \rho^4 \\ \rho & 1 & \rho & \rho^2 & \rho^3 \\ \rho^2 & \rho & 1 & \rho & \rho^2 \\ \rho^3 & \rho^2 & \rho & 1 & \rho \\ \rho^4 & \rho^3 & \rho^2 & \rho & 1 \end{pmatrix}$	2
Unstructured (UN)	$\begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \sigma_{14} & \sigma_{15} \\ \sigma_{12} & \sigma_2^2 & \sigma_{23} & \sigma_{24} & \sigma_{25} \\ \sigma_{13} & \sigma_{23} & \sigma_3^2 & \sigma_{34} & \sigma_{35} \\ \sigma_{14} & \sigma_{24} & \sigma_{34} & \sigma_4^2 & \sigma_{45} \\ \sigma_{15} & \sigma_{25} & \sigma_{35} & \sigma_{45} & \sigma_5^2 \end{pmatrix}$	15

Table 15. Heterogeneous versions of the structures (variances σ_j^2 free at each occasion; five occasions). The number of parameters, not given on the slide, is 6 for CSH and ARH(1) and 9 for TOEPH (variances plus correlation parameters).

Structure	Example
Heterogeneous compound symmetry (CSH)	$\begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho & \sigma_1\sigma_3\rho & \sigma_1\sigma_4\rho & \sigma_1\sigma_5\rho \\ \sigma_1\sigma_2\rho & \sigma_2^2 & \sigma_2\sigma_3\rho & \sigma_2\sigma_4\rho & \sigma_2\sigma_5\rho \\ \sigma_1\sigma_3\rho & \sigma_2\sigma_3\rho & \sigma_3^2 & \sigma_3\sigma_4\rho & \sigma_3\sigma_5\rho \\ \sigma_1\sigma_4\rho & \sigma_2\sigma_4\rho & \sigma_3\sigma_4\rho & \sigma_4^2 & \sigma_4\sigma_5\rho \\ \sigma_1\sigma_5\rho & \sigma_2\sigma_5\rho & \sigma_3\sigma_5\rho & \sigma_4\sigma_5\rho & \sigma_5^2 \end{pmatrix}$
Heterogeneous first-order autoregressive [ARH(1)]	$\begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho & \sigma_1\sigma_3\rho^2 & \sigma_1\sigma_4\rho^3 & \sigma_1\sigma_5\rho^4 \\ \sigma_1\sigma_2\rho & \sigma_2^2 & \sigma_2\sigma_3\rho & \sigma_2\sigma_4\rho^2 & \sigma_2\sigma_5\rho^3 \\ \sigma_1\sigma_3\rho^2 & \sigma_2\sigma_3\rho & \sigma_3^2 & \sigma_3\sigma_4\rho & \sigma_3\sigma_5\rho^2 \\ \sigma_1\sigma_4\rho^3 & \sigma_2\sigma_4\rho^2 & \sigma_3\sigma_4\rho & \sigma_4^2 & \sigma_4\sigma_5\rho \\ \sigma_1\sigma_5\rho^4 & \sigma_2\sigma_5\rho^3 & \sigma_3\sigma_5\rho^2 & \sigma_4\sigma_5\rho & \sigma_5^2 \end{pmatrix}$
Heterogeneous Toeplitz (TOEPH)	$\begin{pmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho_1 & \sigma_1\sigma_3\rho_2 & \sigma_1\sigma_4\rho_3 & \sigma_1\sigma_5\rho_4 \\ \sigma_1\sigma_2\rho_1 & \sigma_2^2 & \sigma_2\sigma_3\rho_1 & \sigma_2\sigma_4\rho_2 & \sigma_2\sigma_5\rho_3 \\ \sigma_1\sigma_3\rho_2 & \sigma_2\sigma_3\rho_1 & \sigma_3^2 & \sigma_3\sigma_4\rho_1 & \sigma_3\sigma_5\rho_2 \\ \sigma_1\sigma_4\rho_3 & \sigma_2\sigma_4\rho_2 & \sigma_3\sigma_4\rho_1 & \sigma_4^2 & \sigma_4\sigma_5\rho_1 \\ \sigma_1\sigma_5\rho_4 & \sigma_2\sigma_5\rho_3 & \sigma_3\sigma_5\rho_2 & \sigma_4\sigma_5\rho_1 & \sigma_5^2 \end{pmatrix}$

Pinheiro and Bates (2000, *Mixed Effects Models in S and S-Plus*), whose tables of covariance structures are reproduced above, illustrate the `nlme` classes with the `Orthodont` data. To specify an AR(1) correlation structure with $\phi = 0.8$, with the position variable given by the observation order within group and the grouping variable `Subject`, they use:

```
> cs1AR1 <- corAR1( 0.8, form = ~ 1 | Subject )
> cs1AR1 <- initialize( cs1AR1, data = Orthodont )
> corMatrix( cs1AR1 )
$M01:
      [,1] [,2] [,3]  [,4]
[1,] 1.000 0.80 0.64 0.512
[2,] 0.800 1.00 0.80 0.640
[3,] 0.640 0.80 1.00 0.800
[4,] 0.512 0.64 0.80 1.000
```

As described in §5.3.1 of that book, the AR(1) model is equivalent to an ARMA(1, 0) model, so that the `corARMA` class can also be used to represent a `corAR1` object; however, the `corAR1` methods are designed to take advantage of the particular structure of the AR(1) model and are substantially more efficient than the corresponding `corARMA` methods.

Choice among covariance pattern models

How to choose the most appropriate covariance model, and how misspecification affects inference? The model for the covariance depends on the model for the mean, since it tries to account for the covariance among residuals $(y_{is} - \mu_{is}(\beta))(y_{ik} - \mu_{ik}(\beta))$. General guidelines: first find a *maximal* model for the mean, then decide on the covariance. This is easy if there are not many variables or interactions, and is based on subject-matter considerations. Maximal models typically include treatment covariates (RCT) or quasi-treatment covariates (exposure groups in observational studies) and their interaction with time. Whether to include additional variables or interactions must be decided on subject-matter grounds, *not through an automatic procedure*. Given a maximal mean model, a sequence of covariance-pattern models can be fitted; when two models are nested (e.g. compound symmetry within Toeplitz) a REML likelihood-ratio test can compare the full and reduced covariance models. (The slides also show Pinheiro and Bates (2000, *Mixed Effects Models in S and S-Plus*) figures of covariance functions.)

Likelihood inference for linear mixed models

Standard estimation methods for linear (better, Gaussian) mixed models are maximum likelihood (ML) and restricted maximum likelihood (REML). Likelihood inference requires distributional assumptions on the random and error terms: $\varepsilon_i \mid \sigma^2 \overset{\text{iid}}{\sim} N_{n_i}(0, E_i)$ and $b_i \mid D \overset{\text{iid}}{\sim} N_{q+1}(0, D)$, with $E_i = \sigma^2 I_{n_i}$ and D unstructured,

$$D = \begin{pmatrix} \sigma_0^2 & \sigma_{01} & \cdots & \sigma_{0q} \\ \sigma_{10} & \sigma_1^2 & \cdots & \sigma_{1q} \\ \vdots & & \ddots & \vdots \\ \sigma_{q0} & \sigma_{q1} & \cdots & \sigma_q^2 \end{pmatrix}.$$

Hence the variance–covariance parameter vector is $\alpha = [\sigma^2, D]$. In the frequentist domain the random part models within-individual correlations, serial correlations, and so on; frequentist inference focuses primarily on the fixed-effect parameters β and the variance components α, though one may also be interested in $b = [b_1, \ldots, b_m]^\top$. We are often interested in the marginal mean and variance, $\mathbb{E}[Y_i \mid \beta] = \mu_i(\beta) = x_i\beta$ and $\mathbf{var}(Y_i \mid \alpha) = V_i(\alpha) = z_i D z_i^\top + \sigma^2 I_{n_i}$.

To conduct inference on β and α we integrate over the random effects in the two-stage model,

$$p(y \mid \beta, \alpha) = \int p(y \mid b, \beta, \alpha)\, p(b \mid \beta, \alpha)\, db = \prod_{i=1}^{m} \int p(y_i \mid b_i, \beta, \sigma^2)\, p(b_i \mid D)\, db_i,$$

and since a convolution of normal distributions is normal, $y_i \mid \beta, \alpha \sim N_{n_i}[\mu_i(\beta), V_i(\alpha)]$. The log-likelihood is

$$\ell(\beta, \alpha) = -\frac{1}{2}\sum_{i=1}^{m} \log |V_i(\alpha)| - \frac{1}{2}\sum_{i=1}^{m} (y_i - x_i\beta)^\top V_i(\alpha)^{-1}(y_i - x_i\beta),$$

and the score functions are

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^{m} x_i^\top V_i^{-1}(Y_i - x_i\beta), \qquad \frac{\partial \ell}{\partial \alpha_r} = \frac{1}{2}\Big\{ (y - x_i\beta)' V_i^{-1} \frac{\partial V_i}{\partial \alpha_r} V_i^{-1} (y - x_i\beta) - \operatorname{tr}\Big(V_i^{-1} \frac{\partial V_i}{\partial \alpha_r} \Big) \Big\}.$$

The MLE of β is the *generalised least squares* (GLS) estimator

$$\hat\beta = \Big(\sum_{i=1}^{m} x_i^\top V_i(\hat\alpha)^{-1} x_i\Big)^{-1} \Big(\sum_{i=1}^{m} x_i^\top V_i(\hat\alpha)^{-1} y_i\Big),$$

where $V_i(\hat\alpha)$ is a plug-in based on an estimate of the variance components. If $D = 0$ then $V = \sigma^2 I_N$, $N = \sum_i m_i$, and $\hat\beta$ is the ordinary least squares estimator. It is easy to prove that $\operatorname{var}(\hat\beta) = \big(\sum_i x_i^\top V_i(\alpha)^{-1} x_i\big)^{-1}$.

Remarks. The result can be seen as the maximisation of the profile likelihood of β given $V(\alpha)$ or an estimate thereof. The expected information matrix is block diagonal, $I(\beta, \alpha) = \operatorname{diag}(I_{\beta\beta}, I_{\alpha\alpha})$, so $\hat\beta$ and $\hat\alpha$ are asymptotically independent and any consistent estimator of α gives an asymptotically efficient estimator of β. $\hat\beta$ is linear in the data Y_i, hence normal under normality. Under correct specification of the variance model and with a consistent $\hat\alpha$,

$$\Big(\sum_{i=1}^{m} x_i V_i(\hat\alpha)^{-1} x_i\Big)^{1/2} (\hat\beta_m - \beta) \xrightarrow{d} N_{k+1}(0, I) \quad \text{as } m \to \infty.$$

Since $\hat\beta$ is linear in Y, these properties hold for large samples even if the sampling distribution of Y_i is not multivariate normal (Laird and Ware, 1982); the second moments of the data must be correctly specified.

Optimisation

The maximisation follows the standard procedures for general linear models: the profiled log-likelihood is optimised through EM iterations or Newton–Raphson iterations (Laird and Ware, 1982).

- **EM algorithm** (Dempster, Laird and Rubin, 1977; see Part I): the random effects b_i are regarded as unobserved data. Starting from guesses of α, β, iterate: (1) find the distribution of $b \mid y$ under the current parameter estimates; (2) treating that distribution as fixed (not depending on α, β), write $Q(\alpha, \beta) = \mathbb{E}_{\cdot \mid y}\{\log f(y, b \mid \beta)\}$ as a function of α, β, with y fixed (E-step); (3) maximise Q with respect to the parameters to obtain updated $\hat\alpha, \hat\beta$ (M-step). The expectation in step 2 is taken with respect to the fixed distribution from step 1. Each iteration increases the log-likelihood until convergence.
- **Newton–Raphson** (Thisted, 1988): uses a first-order expansion of the score around the current estimate $\alpha^{(w)}$ to produce $\alpha^{(w+1)}$; each iteration needs the score and the Hessian of the log-likelihood. Under general conditions it converges quadratically. As the Hessian may be expensive, quicker approximations lead to quasi-Newton algorithms.
- **Trade-off.** EM iterations are quickly computed and bring the parameters into the region of the optimum very quickly, but progress is slow near the optimum. Newton–Raphson iterations are individually more expensive and can be unstable far from the optimum, but converge very quickly close to it. A *hybrid* approach starts from $\alpha^{(0)}$, performs a moderate number of EM iterations, then switches to Newton–Raphson. The `lme` function of the `nlme` package implements this scheme: it calculates initial estimates, performs by default 25 EM iterations and then switches to Newton–Raphson.

Testing fixed effects

Tests of fixed effects are typically Wald or likelihood-ratio (LRT) tests. With asymptotic distributions and independent predictors they are equivalent; with small samples or correlated predictors they can vary considerably. Let $\tilde L$ be a design vector or matrix of known weights selecting components of β

(weight 1 for the components selected, 0 otherwise) and $\tilde{L}\beta$ a combination of interest. The sampling distribution of $\tilde{L}\hat{\beta}$ is multivariate normal with mean $\tilde{L}\beta$ and covariance

$$\operatorname{cov}(\tilde{L}\hat{\beta}) = \tilde{L}\operatorname{cov}(\hat{\beta})\tilde{L}' = \tilde{L}\Big(\sum_{i=1}^{N} X_i'\hat{V}_i^{-1}X_i\Big)^{-1}\tilde{L}'.$$

Wald test. $H_0 : \tilde{L}\beta = 0$ versus $H_A : \tilde{L}\beta \neq 0$ is tested with

$$W^2 = (\tilde{L}\hat{\beta})'\Big\{\tilde{L}\Big(\sum_{i=1}^{N} X_i'\hat{V}_i^{-1}X_i\Big)^{-1}\tilde{L}'\Big\}^{-1}(\tilde{L}\hat{\beta}),$$

asymptotically χ^2 with $\operatorname{rank}(\tilde{L})$ degrees of freedom. Similarly the approximate confidence interval, given α, is $\tilde{L}\hat{\beta} \pm t_{df,\alpha/2} \times \big[\tilde{L}(\sum_i X_i'\hat{V}_i^{-1}X_i)^{-1}\tilde{L}'\big]^{1/2}$. For a single component a Z-test works as an approximate Wald test.

Bias of the Wald statistic. The Wald statistic is considered biased downward because the variability in estimating the variance components is not taken into account in the ML estimates (Dempster et al., 1981). The bias can be reduced with an approximate F-statistic,

$$F = \frac{(\tilde{L}\hat{\beta})'\big\{\tilde{L}\big[(\sum_{i=1}^{N} X_i'\hat{V}_i^{-1}(\hat{\alpha})X_i)^{-1}\big]\tilde{L}'\big\}^{-1}(\tilde{L}\hat{\beta})}{\operatorname{rank}(\tilde{L})},$$

with $\operatorname{rank}(\tilde{L})$ numerator degrees of freedom and denominator degrees of freedom that must be estimated from the data (several methods, omitted); this uncertainty somewhat restricts the use of the F-test.

Likelihood-ratio test. To test $H_0 : \tilde{L}\beta = 0$ against $H_A : \tilde{L}\beta \neq 0$, compare the maximised log-likelihoods of two models,

$$G^2 = 2\log L(\hat{\theta}_{\text{full}}) - 2\log L(\hat{\theta}_{\text{reduced}}),$$

asymptotically χ^2 with degrees of freedom equal to the difference in the number of fixed-effects parameters. If G^2 has a p-value smaller than α, the null hypothesis about θ is rejected.

Part IV

Bayesian Statistics: Models, Inference and Computation

Lecture 3 of 3 · F. Ungolo, UNSW Sydney

Frequentist versus Bayesian inference

Frequentist inference. A random variable Y is drawn through a probability model $f(y \mid \theta)$ with indexing parameter θ, which is *fixed*. A sample $y = (y_1, \ldots, y_n)$ is observed and used to estimate the unknown θ through the likelihood $L(\theta \mid y)$, which also allows the sampling uncertainty to be assessed.

Bayesian inference. The parameter θ is also a random variable, with probability distribution $\pi(\theta)$ chosen on the basis of previous studies, expert knowledge and the problem at hand. The likelihood is used to update $\pi(\theta)$ and obtain the posterior distribution

$$\pi(\theta \mid y) \propto L(\theta \mid y)\, \pi(\theta).$$

The posterior characterises the uncertainty about θ, supports prediction for future observations and model checking.

Learning objectives

By the end of the lecture, students should be able to:

- formulate a likelihood and a prior, and interpret the posterior and its summaries;
- derive and report a one-parameter Beta–Binomial analysis;
- explain joint and marginal posteriors in a multiparameter model;
- describe the computational aspects of Bayesian analysis and MCMC methods: Metropolis–Hastings, Gibbs, Hamiltonian Monte Carlo.

Application: estimating an admission probability

Consider women applying to the Berkeley graduate department coded A in fall 1973. Let Y be the indicator equal to 1 in case of admission and 0 otherwise; $Y \sim \text{Bernoulli}(p)$, where p is the probability of admission. The data will be the observed indicators; the sum z of the indicators over a sample of size n is a realisation of a Binomial(n, p) distribution.

- **Frequentist analysis:** p is fixed but unknown; given n observations, estimate $\hat{p} = z/n$.
- **Bayesian analysis:** before observing the data assign the prior $p \sim \text{Beta}(1, 1)$ and update it with the data to obtain $\pi(p \mid y)$. The uniform prior expresses lack of information about p; in applied analyses we would examine whether it expresses reasonable knowledge before fitting.

The public R object `datasets::UCBAdmissions` gives admitted and total applications for each gender and department (letters A–F are codes in the released R table, a total of 4,526 applications; source: R documentation and Bickel, Hammel and O'Connell, 1975).

Table 16. Berkeley admissions: observed counts by department and gender.

Department code	Men		Women	
	Admitted	Total	Admitted	Total
A	512	825	89	108
B	353	560	17	25
C	120	325	202	593
D	138	417	131	375
E	53	191	94	393
F	22	373	24	341
All six	1,198	2,691	557	1,835

For women applying to Department A: $z = 89$ admitted out of $n = 108$, $\hat{p} = z/n = 0.824$. Our inferential question is how uncertain we are about p after observing these counts.

How does Bayesian data analysis inform the researcher?

Suppose we have counts of applications and offers in each department. We want to understand the admission pattern and anticipate future outcomes.

1. **Describe uncertainty before seeing the counts.** Assign plausible ranges to the unknown admission probabilities and make explicit what is being assumed.
2. **Learn from the observed counts.** Update these ranges and ask how likely it is that women have lower admission odds than men applying to the same department.
3. **Check and predict.** Ask whether the fitted model reproduces admissions in each group; predict the number of offers among future applicants with a specified profile.

Practical value: the answer includes both an estimate and a measure of uncertainty, with assumptions that we can inspect and revise.

Bayes' rule, conjugacy and the Beta–Binomial model

Posterior distribution and Bayes' rule

Let $Y = (Y_1, \ldots, Y_n)$ be i.i.d. random variables with observed values $y = (y_1, \ldots, y_n)$, and let $\theta \in \Theta$ be the unknown parameter vector indexing the distribution of Y_i: $f(y \mid \theta) = \prod_{i=1}^{n} f(y_i \mid \theta)$. The posterior distribution is

$$\pi(\theta \mid y) = \frac{f(y \mid \theta)\, \pi(\theta)}{f(y)}, \qquad f(y) = \int_{\Theta} f(y \mid \theta)\, \pi(\theta)\, d\theta,$$

where $f(y)$ is the distribution of y marginalised with respect to the "latent" θ. For fixed observed y, $f(y)$ is the normalising constant that makes the posterior integrate (or sum, if discrete) to one.

For the Department A example, with p the admission probability and $z = 89$ offers among $n = 108$ applicants, begin with $p \sim \text{Beta}(1, 1)$:

$$\underbrace{\pi(p \mid y)}_{\text{posterior}} \propto \underbrace{L(p \mid y)}_{\text{likelihood}} \underbrace{\pi(p)}_{\text{prior}}.$$

The likelihood is rescaled: it is not itself a probability distribution over p. With a uniform prior its rescaled shape coincides with the posterior.

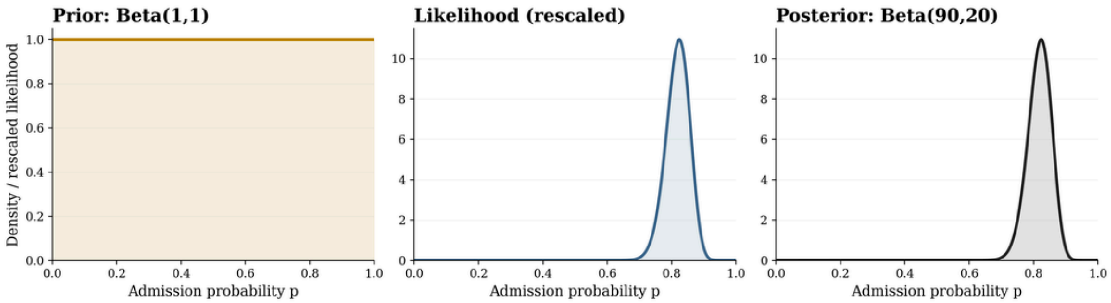


Figure 11. Prior Beta(1, 1), rescaled likelihood and posterior Beta(90, 20) for the admission probability of women in Department A. With a uniform prior the rescaled likelihood and the posterior have the same shape.

Worked update and the Beta posterior

Among $n = 108$ women applying to A, $\sum_{i=1}^{108} y_i = 89$ were admitted and 19 were not. The Bernoulli likelihood and a Beta$(1, 1)$ prior give

$$\pi(p \mid y) \propto \underbrace{p^{89}(1-p)^{19}}_{\text{Binomial lik.}} \underbrace{p^{1-1}(1-p)^{1-1}}_{\text{Beta prior kernel}} \propto p^{90-1}(1-p)^{20-1},$$

the kernel of a Beta density with parameters 90 and 20: $p \mid y \sim \text{Beta}(90, 20)$. The Beta distribution is *conjugate* for the parameter p of a Bernoulli (and also Binomial) distribution: the posterior is of the same parametric family as the prior. In general, for a Beta(a, b) prior,

$$\pi(p \mid y) \propto \underbrace{p^{z}(1-p)^{n-z}}_{\text{likelihood kernel}} \underbrace{p^{a-1}(1-p)^{b-1}}_{\text{prior kernel}} \propto p^{a+z-1}(1-p)^{b+n-z-1}, \qquad p \mid z \sim \text{Beta}(a+z,\, b+n-z).$$

The posterior parameters add the observed admissions and rejections to the prior parameters.

Convenience of conjugate priors

For a specified likelihood, a prior family is conjugate when the posterior belongs to the same family: Beta(a, b) with a Binomial likelihood (y admissions in n applications) gives Beta$(a + y, b + n - y)$. The posterior has a closed form, so posterior means, intervals and predictions can be calculated algebraically. In a multiparameter model the joint posterior may not be available in closed form; nevertheless, for some parameters the *conditional* posterior may have a standard form. This is *conditional conjugacy* and can enhance the computation of the posterior. Computational convenience is useful, but it cannot substitute for a plausible prior or a check of its implications.

Reading a posterior distribution

The Berkeley example gives $p \mid y \sim \text{Beta}(90, 20)$ under a uniform prior (figure below): the posterior mean is $\mathbb{E}(p \mid y) = 0.818$; the central shaded area has posterior probability 0.95; and $P(p > 0.80 \mid y) = 0.703$. Predicting new admissions also requires allowing for Binomial variation in future outcomes.

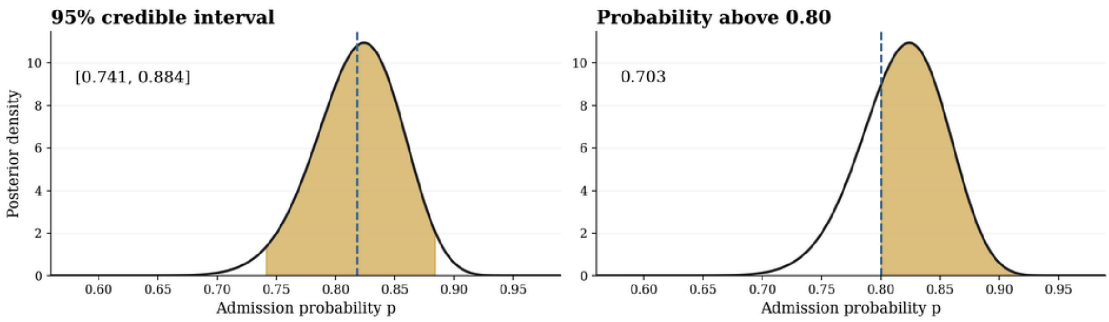


Figure 12. The posterior Beta(90, 20). Left: the 95% equal-tail credible interval $[0.741, 0.884]$; the dashed line is the posterior mean 0.818. Right: the shaded area is $P(p > 0.80 \mid y) = 0.703$.

How would we choose a prior for admissions?

For a Beta(a, b) prior on p, $a/(a+b)$ is its mean and $a+b$ controls how concentrated it is around that mean.

- **Previous analyses:** suppose 72 of 100 applicants in a separate, comparable earlier cohort were admitted. Updating Beta$(1, 1)$ with those counts gives Beta$(73, 29)$ as a possible prior for the new cohort.
- **Expert assessment:** if an expert expects about 70% admission but is quite uncertain, Beta$(7, 3)$ has mean 0.70 and moderate concentration; these shape parameters are an elicitation.
- **Noninformative:** Beta$(1, 1)$ is uniform on $p \in [0, 1]$.
- **Prior–posterior sensitivity:** first inspect the number of events each prior predicts, then examine how the posterior changes under different priors.

Prior predictive checks

Before updating with the observed count, simulate from the proposed model, $p^{(s)} \sim \pi(p) = \text{Beta}(1, 1)$, $Z^{(s)} \mid p^{(s)} \sim \text{Binomial}(108, p^{(s)})$, with $Z^{(s)} = \sum_{i=1}^{108} Y_i^{(s)}$. Based on the sampled values of p, and thus of Z, assess whether the values of Z are plausible. If not, reconsider the prior before using the observed count $y = 89$.

The posterior mean as a weighted average

For $p \sim \text{Beta}(a, b)$ and $Z \mid p \sim \text{Binomial}(n, p)$,

$$\mathbb{E}(p \mid z) = \frac{a+z}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \frac{z}{n}.$$

The prior mean is $a/(a+b)$ and the observed proportion z/n; the weights reflect the prior concentration $a+b$ and the sample size n. For $a = b = 1$, $z = 89$, $n = 108$ the weights are $2/110$ and $108/110$, which explains why the posterior mean is close to, but not exactly, $89/108$.

When does the likelihood outweigh the prior?

Keep a Beta$(10, 10)$ prior (mean 0.5, concentration 20) and compare two samples with an observed admission rate of about 82.4%:

Table 17. The same fixed prior has less influence as n grows.

Sample	Prior weight $20/(20+n)$	Posterior	Posterior mean
Berkeley: 89/108	15.6%	Beta(99, 29)	0.773
Hypothetical: 890/1080	1.8%	Beta(900, 200)	0.818

The posterior mean approaches the observed proportion; thus a conjugate prior may be chosen partly for computational convenience, especially when large samples are available.

Summary of the worked posterior and prior sensitivity

For $p \mid z \sim \text{Beta}(90, 20)$: observed proportion $89/108 = 0.824$; posterior mean $90/110 = 0.818$; 95% equal-tail credible interval $[0.741,\ 0.884]$; posterior probability $P(p > 0.80 \mid z) = 0.703$. Both priors considered have mean 0.5 but express different degrees of prior certainty: $\text{Beta}(1, 1) \to \text{Beta}(90, 20)$, posterior mean 0.818; $\text{Beta}(10, 10) \to \text{Beta}(99, 29)$, posterior mean 0.773. The stronger prior pulls the posterior further toward 0.5 despite the same data.

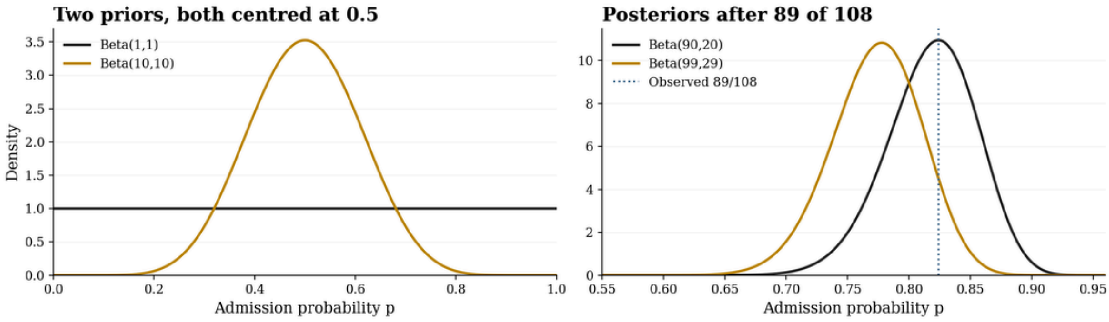


Figure 13. Prior–posterior sensitivity. Left: two priors with mean 0.5, Beta(1, 1) and Beta(10, 10). Right: the corresponding posteriors, Beta(90, 20) and Beta(99, 29), with the observed proportion $89/108$ (dotted line).

From estimation to prediction

For m new applicants in the same setting, the predicted value of Z_{new} is its expected value given the observed z, averaged over the uncertainty in p:

$$\mathbb{E}[Z_{\text{new}} \mid z] = \sum_{j=0}^{m} j \int \underbrace{\binom{m}{j} p^j (1-p)^{m-j}}_{\text{p.m.f. Binomial}(m,p)} \underbrace{\pi(p \mid z)}_{\text{posterior of } p} \, dp.$$

For $m = 10$ and $p \mid y \sim \text{Beta}(90, 20)$: $\mathbb{E}(Z_{\text{new}} \mid z) = 10(0.818) = 8.18$, with 95% predictive interval $[5, 10]$. The distribution of future counts includes both the uncertainty about p and the Binomial variation in new outcomes.

```
# Summarise Beta(90, 20) without MCMC.
qbeta(c(0.025, 0.5, 0.975), 90, 20)
1 - pbeta(0.8, 90, 20)                  # Posterior Pr(p > 0.8 | y).
# Exact predictive quantiles for 10 new applicants.
pred_pmf <- sapply(0:10, function(k)
  choose(10, k) * beta(90 + k, 20 + 10 - k) / beta(90, 20))
cdf <- cumsum(pred_pmf)
c(lower = which(cdf >= 0.025)[1] - 1,
  upper = which(cdf >= 0.975)[1] - 1)
```

Output (rounded): posterior quantiles $0.741,\ 0.820,\ 0.884$; $P(p > 0.8 \mid y) = 0.703$; predictive interval $[5, 10]$.

A model for admission by sex and department

The Beta–Binomial analysis focused on women applying to Department A. Suppose now we analyse the admission pattern at Berkeley by sex *and* department; sex and department yield 12 combinations. The overall admission proportions are 44.5% for men and 30.4% for women; the applicant mix differs by gender (the A–F labels are codes in the R dataset; the academic fields are not identified).

Table 18. Overall and department-specific admission rates (admitted applicants divided by all applicants of that gender in that department).

Department	Men admitted (%)	Women admitted (%)
A	62.1	82.4
B	63.0	68.0
C	36.9	34.1
D	33.1	34.9
E	27.7	23.9
F	5.9	7.0
All six departments combined	44.5	30.4

The combined percentages pool across the six departments: $1{,}198/2{,}691 = 44.5\%$ for men and $557/1{,}835 = 30.4\%$ for women. Women have a higher rate in four of six departments but a lower combined rate: the mix of departments applied to differs (an instance of Simpson's paradox).

Previously: one model for each combination

Six departments and two genders give 12 groups. For applicant i in group k let $y_{i,k}$ be the indicator of a successful application, with success probability p_k: $Y_{i,k} \mid p_k \sim \text{Bernoulli}(p_k)$, $k = 1, \ldots, 12$. With independent outcomes, identically distributed by sex and department,

$$L(p_k \mid y) = \prod_{i=1}^{n_k} p_k^{y_{i,k}} (1 - p_k)^{1 - y_{i,k}}, \qquad i = 1, \ldots, n_k;\ k = 1, \ldots, 12.$$

Each p_k is estimated from its own group (for example, B/women has only 25 applicants). These separate analyses do not allow statistical information to be shared across groups.

A Berkeley admission regression model

For applicant i, let $Y_i = 1$ if admitted and 0 otherwise, $F_i = 1$ for women and 0 for men, and d_i the department applied for:

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \alpha + \beta F_i + \gamma_{d_i}, \qquad \gamma_A = 0.$$

α is the log-odds for men in A; γ_d shifts the log-odds of department d relative to A; β is the regression coefficient for the effect of sex on the logit admission rate, all else being equal. The following illustrative priors are assumed:

$$\alpha \sim N(0, 2.5^2), \quad \beta \sim N(0, 2.5^2), \quad \gamma_d \sim N(0, 2.5^2),\ d = B, C, D, E, F.$$

The female–male odds ratio is $\text{OR}_F = \exp(\alpha + \beta)/\exp(\alpha) = \exp(\beta)$: $\text{OR}_F < 1$ means lower admission odds for women under this model, $\text{OR}_F > 1$ higher. Before fitting, check the admission probabilities and rates implied by the priors.

Predicted values under the prior

The inverse logit converts log-odds w into a probability, $p = \text{logit}^{-1}(w) = 1/(1 + e^{-w})$. Under the independent $N(0, 2.5^2)$ priors consider $p_{A,M} = \text{logit}^{-1}(\alpha)$ and $p_{C,F} = \text{logit}^{-1}(\alpha + \beta + \gamma_C)$:

Table 19. Central 95% prior range for the admission probability.

Group	Prior SD of log-odds	Central 95% prior range for p
Department A / male	2.5	[0.007, 0.993]
Department C / female	$2.5\sqrt{3}$	[0.0002, 0.9998]

The central 95% range lies between the 2.5th and 97.5th prior percentiles of p; for normal log-odds with SD s its endpoints are $[\text{logit}^{-1}(-1.96s), \text{logit}^{-1}(1.96s)]$.

The posterior distribution of the regression model

The model has parameters $\theta = (\alpha, \beta, \gamma_B, \ldots, \gamma_F)$. The likelihood kernel (the omitted Binomial factors $\prod_k \binom{n_k}{y_k}$ and the normalising constant do not depend on θ; they change the overall scale, not the relative posterior density) is

$$L(\alpha, \beta, \gamma_B, \ldots, \gamma_F \mid y) = \prod_{i=1}^{n} p_i(\theta)^{y_i} \{1 - p_i(\theta)\}^{1 - y_i}.$$

Independent priors then give the joint posterior

$$\pi(\theta \mid y) = \frac{L(\theta \mid y)\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d)}{\underbrace{\int L(\theta \mid y)\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d)\,d\theta}_{f(y)}} \propto L(\theta \mid y)\,\pi(\alpha)\pi(\beta)\prod_{d=B}^{F}\pi(\gamma_d).$$

Marginalisation: averaging over the other parameters

Suppose the research question concerns β. Its marginal posterior is

$$\pi(\beta \mid y) = \int \pi(\alpha, \beta, \gamma \mid y)\, d\alpha\, d\gamma.$$

For each candidate β we integrate over α and γ, weighted by the joint posterior density. Uncertainty about the department effects is retained in this integral; plugging in one estimate for each γ_d would discard it. Similarly we can obtain a marginal for α, any γ_d, or a derived group probability p_k.

Preparing the grouped data in R

```
# Convert the built-in three-way table to one row per gender and department.
counts <- as.data.frame(datasets::UCBAdmissions)
# Put admitted and rejected counts into separate columns.
wide <- reshape(counts, idvar = c("Gender", "Dept"),
                timevar = "Admit", direction = "wide")
names(wide) <- sub("^Freq\\.", "", names(wide))
names(wide)[names(wide) == "Admitted"] <- "admitted"
names(wide)[names(wide) == "Rejected"] <- "rejected"
# Create the binomial denominator and model covariates.
wide$n <- wide$admitted + wide$rejected
wide$female <- as.integer(wide$Gender == "Female")
wide$dept_id <- as.integer(factor(wide$Dept, levels = LETTERS[1:6]))
stopifnot(sum(wide$n) == 4526)
```

What can we report from the joint posterior?

The parameters of interest depend on the research question. The joint posterior supports inference on all seven coefficients and on functions of them.

- If all coefficients matter, report medians and 95% equal-tail intervals for $\alpha, \beta, \gamma_B, \ldots, \gamma_F$, stating that A/men is the reference.

- For the gender comparison, report $\exp(\beta)$ and, if useful, $P(\beta < 0 \mid y)$; exponentiating the equal-tail interval endpoints gives an odds-ratio interval.
- For the effect of the department, report the coefficients γ_d or the corresponding probabilities (by sex) with a posterior interval; for a group k, $p_k = \text{logit}^{-1}(\alpha + \beta F_k + \gamma_{d[k]})$ with a posterior interval.
- In every case, state the priors, the common-effect assumption and the limits of these grouped data.

Markov chain Monte Carlo

Why MCMC?

The Beta–Binomial posterior was available in closed form. For the logistic regression, the seven parameters have normal priors and the joint posterior has no standard closed form; direct integration is complicated, sometimes not even feasible. Markov chain Monte Carlo (MCMC) methods draw sequential samples $\theta^{(1)}, \theta^{(2)}, \ldots, \theta^{(S)}$ whose long-run distribution is the posterior $\pi(\theta \mid y)$ (a law-of-large-numbers argument; the slides loosely describe the draws as independent, but in practice they are dependent and this must be checked). In many models we can evaluate $p(y \mid \theta)\pi(\theta)$ but cannot normalise the posterior or integrate it analytically. A posterior expectation is approximated by an average of draws,

$$\frac{1}{S}\sum_{s=1}^{S} h(\theta^{(s)}) \approx \int h(\theta)\pi(\theta \mid y)\, d\theta.$$

For the Berkeley example, choosing $h(\theta) = 1\{\beta < 0\}$ estimates $P(\beta < 0 \mid y) = \int 1\{\beta < 0\}\pi(\theta \mid y)\, d\theta$ by $\hat{P}(\beta < 0 \mid y) = \frac{1}{S}\sum_s 1\{\beta^{(s)} < 0\}$.

Random sequential draws. Start with a parameter value $\theta^{(0)}$; a sampling rule uses its current value to produce $\theta^{(1)}$, then $\theta^{(1)}$ to produce $\theta^{(2)}$, and so on: $\theta^{(0)} \to \theta^{(1)} \to \theta^{(2)} \to \cdots$. For instance, a proposed move from an admission probability of 0.78 might be accepted as 0.82 or rejected, leaving the next state at 0.78. Each draw depends on the preceding state; the sequence is not a set of independent draws. Design the rule so that its long-run distribution is the posterior; discard initial warm-up draws and check that multiple chains explore it well.

Metropolis–Hastings: propose, then accept or retain

Let $\tilde{\pi}(\theta \mid y) = f(y \mid \theta)\pi(\theta)$ be the unnormalised posterior. At iteration s:

1. Draw θ^* from a proposal distribution, $\theta^* \sim q(\theta^* \mid \theta^{(s)})$.
2. Accept θ^* with probability

$$a = \min\left\{1, \frac{\tilde{\pi}(\theta^* \mid y)\, q(\theta^{(s)} \mid \theta^*)}{\tilde{\pi}(\theta^{(s)} \mid y)\, q(\theta^* \mid \theta^{(s)})}\right\}.$$

3. Otherwise retain $\theta^{(s)}$; a rejected move still counts as an iteration.

The unknown normalising constant cancels.

For the Berkeley model, with $\theta = (\alpha, \beta, \gamma_B, \ldots, \gamma_F)$, the log-posterior kernel is

$$\ell(\theta) = \log \pi(\theta \mid y) = \underbrace{\sum_{i=1}^{n}\left[y_i \log p_i(\theta) + (1 - y_i)\log\{1 - p_i(\theta)\}\right]}_{\text{Bernoulli log-likelihood}} \underbrace{-\frac{1}{2(2.5)^2}\sum_{j=1}^{7}\theta_j^2}_{\text{Normal log prior}} + C.$$

At each iteration propose a joint move in all seven coefficients, compare $\ell(\theta^*)$ with $\ell(\theta)$, and accept or retain the current value; the normalising constant C cancels. Run four chains from different starting values, discard warm-up and inspect their traces before interpreting summaries.

R: target density, proposal and four Metropolis chains

```
# One row per observed department/gender cell, seven coefficient columns.
design <- model.matrix(~ female + Dept, data = wide)
n_par <- ncol(design)
# Log likelihood plus independent N(0, 2.5^2) log priors.
log_post <- function(theta) {
  eta <- drop(design %*% theta)
  sum(dbinom(wide$admitted, wide$n, plogis(eta), log = TRUE)) +
    sum(dnorm(theta, mean = 0, sd = 2.5, log = TRUE))
}
# The classical fit tunes proposals; it is not the Bayesian estimate.
fit_glm <- glm(cbind(admitted, rejected) ~ female + Dept,
               family = binomial(), data = wide)
step_U <- chol(vcov(fit_glm)) * 2.38 / sqrt(n_par)

mh_one <- function(start, iterations = 18000, warmup = 3000) {
  theta <- start; value <- log_post(theta); accepted <- 0
  out <- matrix(NA_real_, iterations, n_par)
  colnames(out) <- colnames(design)
  for (s in seq_len(iterations)) {
    # Propose all coefficients jointly from a symmetric normal random walk.
    proposal <- theta + drop(rnorm(n_par) %*% step_U)
    trial <- log_post(proposal)
    if (log(runif(1)) < min(0, trial - value)) {
      theta <- proposal; value <- trial; accepted <- accepted + 1
    }
    out[s, ] <- theta                 # Rejection repeats the previous state.
  }
  list(draws = out[(warmup + 1):iterations, , drop = FALSE],
       acceptance = accepted / iterations)
}
set.seed(1381)
mh <- lapply(seq_len(4), function(j) mh_one(rnorm(n_par, 0, .2)))
vapply(mh, function(z) z$acceptance, numeric(1))
```

The row vector `rnorm(n_par) %*% step_U` has covariance proportional to `vcov(fit_glm)`; this affects sampling efficiency, while the acceptance rule preserves the target posterior. Each retained row is one joint coefficient draw. The draws depend on their preceding state; mixing needs to be checked.

Inspecting the chains

The traces of β and γ_C (four chains, 1,200 retained iterations shown in the lecture figure) should overlap, fluctuate within a stable range and show no persistent drift. Also check $\hat{R}$ and the effective sample size for every parameter; overlap in one short trace segment is not a proof of convergence.

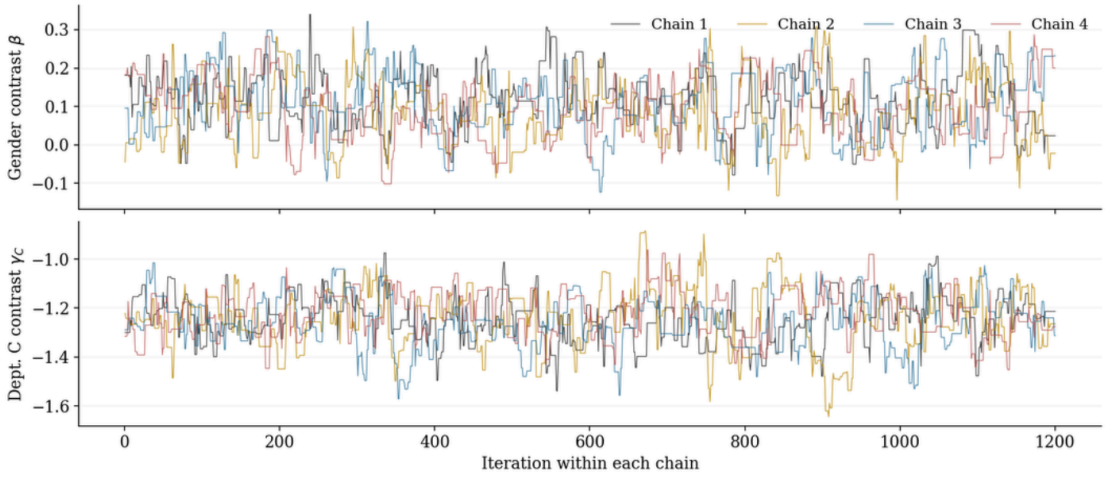


Figure 14. Illustrative four-chain, hand-coded Metropolis run for the Berkeley posterior: traces of the gender contrast β (top) and of the Department C contrast γ_C (bottom), 1,200 retained iterations per chain. The chains overlap, fluctuate within a stable range and show no persistent drift.

A counterexample: chains that do not mix. If the same posterior is targeted but the starting values are dispersed, the proposal steps are very small and only 1,200 iterations are run without warm-up, the chains stay separated: a pooled average or credible interval from this run is unreliable. High acceptance alone does not establish good sampling.

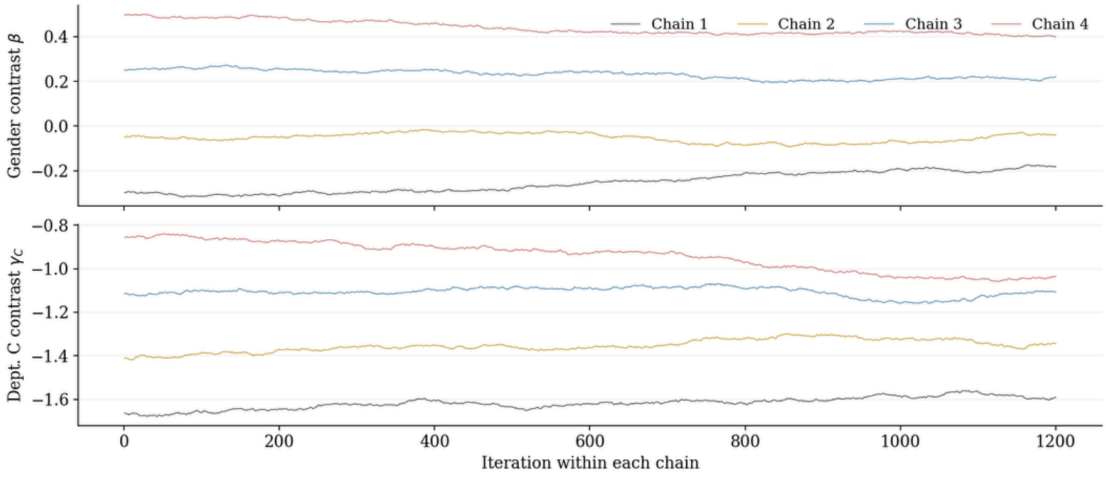


Figure 15. Counterexample: chains that do not mix (dispersed starting values, very small proposal steps, 1,200 iterations without warm-up). The four chains stay separated, so a pooled average or credible interval from this run is unreliable.

Additional checks. Inspect traces for all seven coefficients, looking for separation, slow drift, or long flat stretches from rejected moves. Compare the distributions from the beginning and the end of each retained chain: they should tell a similar story. Examine autocorrelation: when a chain moves slowly, adjacent draws carry nearly the same information. If these checks fail, improve the proposal and run longer chains from dispersed initial values, then repeat the checks before reporting posterior summaries.

Two numerical diagnostics: $\hat{R}$ and ESS

- $\hat{R}$ compares the variation between several chains with the variation within them. Values close to 1 support agreement across chains; substantially larger values signal that the chains have not explored the same distribution.

- The *effective sample size* (ESS) accounts for the dependence between successive draws: it is the approximate number of independent draws with comparable precision for estimating a posterior mean. With four chains of 15,000 retained draws we store 60,000 values per coefficient, but the ESS can be much smaller if successive draws are strongly correlated.

Examine both diagnostics for every coefficient alongside the traces; neither number alone proves convergence or model fit. In this lecture `coda::gelman.diag()` reports a classical potential scale-reduction estimate of $\hat{R}$; Stan later provides modern split and rank-based diagnostics.

```
# Convert four independently run R chains to a coda mcmc.list.
good_coda <- do.call(coda::mcmc.list, lapply(mh, function(z) coda::mcmc(z$draws)))
coda::gelman.diag(good_coda, autoburnin = FALSE)$psrf   # Between- vs within-chain variation per column.
coda::effectiveSize(good_coda)                          # Independent-draw equivalent per coefficient.
# Repeat for deliberately inadequate 1,200-iteration runs.
poor_coda <- do.call(coda::mcmc.list, lapply(mh_poor, function(z) coda::mcmc(z$draws)))
coda::gelman.diag(poor_coda, autoburnin = FALSE)$psrf
coda::effectiveSize(poor_coda)
```

The full script creates `mh_poor` with `step_scale = .02` and dispersed starting values. Do not use those draws for inference.

JAGS: Just Another Gibbs Sampler

JAGS reads a probabilistic model written in BUGS notation: priors, likelihood and named derived quantities. In R, `rjags` passes the model and the data to JAGS, which runs sequential MCMC updates with an update method selected for each part of the model. The output is several chains of posterior draws; `coda` provides summaries, trace plots and convergence diagnostics. From the draws we estimate intervals, probabilities and predictions, including replicated Berkeley tables. For the logistic Berkeley model not every JAGS update is necessarily a direct Gibbs draw; later a normal model shows when full conditionals make Gibbs updates simple. (See the JAGS user manual and the `rjags` reference manual.)

The complete JAGS workflow: Berkeley admissions

1. **Define the question:** estimate admission probabilities and the within-department gender association from the 12 grouped cells.
2. **Specify the model:** Binomial counts, logit probabilities, seven unknown coefficients and their normal priors.
3. **Compute:** start four JAGS chains, adapt and warm up, then retain posterior draws.
4. **Diagnose:** inspect the traces of every coefficient, $\hat{R}$ and the effective sample size.
5. **Infer and check:** summarise marginal posteriors; simulate another 12-cell table and compare it with the observations.
6. **Predict:** use each posterior draw to predict offers among ten future applicants with a specified profile.

```
# JAGS uses precision as the second argument to dnorm.
jags_text <- "
model {
  alpha ~ dnorm(0, prior_precision)
  beta_female ~ dnorm(0, prior_precision)
  beta_dept[1] <- 0
  for (d in 2:6) { beta_dept[d] ~ dnorm(0, prior_precision) }
  for (k in 1:K) {
    admitted[k] ~ dbin(p[k], applicants[k])
    logit(p[k]) <- alpha + beta_female * female[k] + beta_dept[dept[k]]
  }
}
"
# The names below match those in the JAGS model.
jags_data <- list(
  K = nrow(wide),                       # 12 department/gender cells
  admitted = as.integer(wide$admitted), # offers in each cell
```

```
  applicants = as.integer(wide$n),      # applications in each cell
  female = as.integer(wide$female),     # 0 = man; 1 = woman
  dept = as.integer(wide$dept_id),      # A = 1, ..., F = 6
  prior_precision = 1 / 2.5^2           # inverse prior variance
)
```

The zero reference effect identifies Department A; the remaining department coefficients are relative shifts on the log-odds scale. `as.integer()` passes whole-number counts and category codes; the prior $N(0, 2.5^2)$ becomes precision $1/2.5^2$ in JAGS.

```
# Give each chain a distinct, reproducible random-number seed.
inits <- lapply(seq_len(4), function(j)
  list(.RNG.name = "base::Wichmann-Hill", .RNG.seed = 20000L + j))
tc <- textConnection(jags_text)          # Supply the JAGS model text.
jm <- rjags::jags.model(tc, data = jags_data, inits = inits,
                        n.chains = 4, n.adapt = 1000)   # Compile, attach data, adapt four chains.
close(tc)

update(jm, n.iter = 1000)                # Let the chains move before storing draws.
# Monitor exactly seven unknown coefficients; A is the zero reference.
jags_draws <- rjags::coda.samples(jm,
  variable.names = c("alpha", "beta_female", "beta_dept[2:6]"),
  n.iter = 4000)
coef_names <- c("alpha", "beta_female", paste0("beta_dept[", 2:6, "]"))
coef_chains <- jags_draws[, coef_names]
summary(coef_chains)                     # Posterior mean, SD, quantiles by coefficient.

# Seven trace plots (four colours = four chains), then diagnostics.
chain_cols <- c("black", "goldenrod3", "steelblue", "firebrick3")
coda::traceplot(coef_chains, col = chain_cols)
print(coda::gelman.diag(coef_chains, autoburnin = FALSE)$psrf)
print(coda::effectiveSize(coef_chains))
```

`n.adapt` lets JAGS tune its samplers for 1,000 iterations; the next call runs additional warm-up iterations before saving draws. Four chains $\times$ 4,000 retained iterations give 16,000 stored draws per coefficient; diagnose the chains before combining their draws.

What to inspect in the JAGS output. (1) `summary(draws)` gives a mean, SD and quantiles for each monitored variable; for `OR_female` the 2.5% and 97.5% quantiles form an equal-tail credible interval. (2) `plot(draws)` shows trace and density plots; check that the four traces mix and show no persistent drift. (3) Apply `gelman.diag(draws[,coef_names])` and `effectiveSize(draws[,coef_names])`. The output is conditional on the stated model and priors: a well-mixed chain alone does not imply that the logistic model fits the admission counts.

```
# Combine the retained draws from all four chains, after diagnostics.
mat <- do.call(rbind, lapply(jags_draws, as.matrix))
coef_names <- c("alpha", "beta_female", paste0("beta_dept[", 2:6, "]"))
# Median, SD and equal-tail 95% interval per coefficient.
post_summary <- t(vapply(coef_names, function(name) {
  x <- mat[, name]
  c(median = median(x), SD = sd(x),
    low = unname(quantile(x, .025)), high = unname(quantile(x, .975)))
}, numeric(4)))
print(round(post_summary, 2))
```

Posterior summary: all seven coefficients

Table 20. Rounded illustrative four-chain MCMC summaries for the $N(0, 2.5^2)$ priors (repeated runs and JAGS sampling may change the last digit).

Parameter	Median	SD	95% equal-tail credible interval
α	0.57	0.07	[0.44, 0.71]
β	0.09	0.08	[−0.07, 0.25]
γ_B	−0.03	0.11	[−0.24, 0.19]
γ_C	−1.25	0.11	[−1.45, −1.04]
γ_D	−1.28	0.10	[−1.49, −1.08]
γ_E	−1.72	0.13	[−1.97, −1.48]
γ_F	−3.29	0.16	[−3.62, −2.98]

α describes men in A; β is the common gender contrast; each γ_d compares department d with A at the same gender.

Interpreting the posterior coefficients

- The probability of admission for a man applying to Department A is $\text{logit}^{-1}(\alpha)$; α is a baseline log-odds.
- β captures the effect of sex on the log-odds of admission. Draws of $\exp(\beta)$ give a median odds ratio of about 1.10 and a 95% interval $[0.94, 1.29]$.
- γ_C is the Department C–A log-odds difference at the same gender. A value near -1.25 corresponds to odds multiplied by about $\exp(-1.25) = 0.29$.
- The credible intervals describe coefficient uncertainty under a *common* gender effect. Transform each posterior draw to a probability or an odds ratio before calculating an interval for that derived quantity.

Table 21. Reading a posterior summary: Berkeley (rounded summaries from an illustrative MCMC run).

Quantity	Approximate summary	Interpretation
β	median 0.09; 95% CrI [−0.07, 0.25]	adjusted log-odds contrast
$\exp(\beta)$	median 1.10; 95% CrI [0.94, 1.29]	adjusted odds ratio, women versus men
$P(\beta < 0 \mid y)$	0.13	probability of lower adjusted odds

The interval crosses 1 on the odds-ratio scale. The posterior does not establish equal odds; it allows modest differences in either direction.

Sensitivity to the prior distribution

Does the adjusted sex comparison change if each coefficient receives a narrower $N(0, s^2)$ prior? Keeping the same likelihood:

Table 22. Prior-sensitivity of the adjusted sex comparison.

Prior SD s	Median β	Adjusted OR (95% interval)	$P(\beta < 0 \mid y)$
2.5	0.09	1.10 [0.94, 1.29]	0.13
1	0.08	1.08 [0.93, 1.27]	0.16

For this one comparison the inference is similar, though the probability of a negative contrast changes. Sensitivity of one coefficient does not validate the common-effect model; check its predicted cell counts as well. A graphical check compares density estimates of β under the two prior scales (centre, spread and area to the left of zero): the inference changes little, but checking the model's fit requires replicating the observed table.

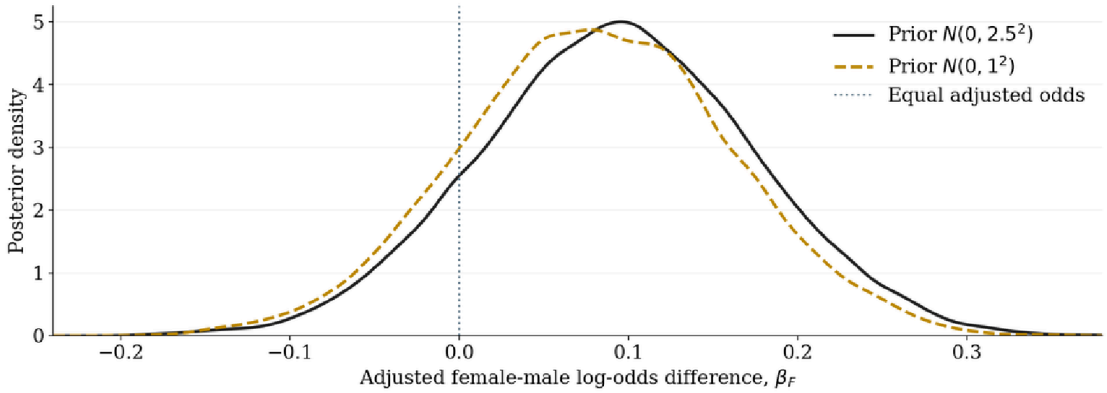


Figure 16. Posterior densities of the adjusted female–male log-odds difference β_F under $N(0, 2.5^2)$ and $N(0, 1^2)$ priors, from illustrative MCMC draws; the dotted line marks equal adjusted odds.

Posterior predictive checks for the Berkeley table

With the chains checked, ask whether the fitted model could produce data like the observed 12-cell table. For every retained draw s and cell k,

$$p_k^{(s)} = \text{logit}^{-1}\big(\alpha^{(s)} + \beta^{(s)} F_k + \gamma_{d[k]}^{(s)}\big), \qquad Y_i^{\text{rep},(s)} \sim \text{Bernoulli}(p_k^{(s)}),$$

a number of draws equal to the total number of applicants of each sex and department yields a new admissions table under the fitted model. Compare the observed z_k with the 2.5th and 97.5th percentiles of its replicated counts, looking for systematic differences by department or gender.

```
# Extract retained cell probabilities from the four MCMC chains.
mat <- do.call(rbind, lapply(jags_draws, as.matrix))
p_names <- paste0("p[", seq_len(nrow(wide)), "]")
p_ppc <- mat[, p_names, drop = FALSE]
set.seed(20261026)                      # Reproduce the Binomial simulations.
# Replicate counts with the original applicant numbers.
y_ppc <- vapply(seq_len(nrow(wide)), function(k)
  rbinom(nrow(p_ppc), size = wide$n[k], prob = p_ppc[, k]), numeric(nrow(p_ppc)))
# Compare the observed counts with their predictive limits.
limits <- apply(y_ppc, 2, quantile, probs = c(0.025, 0.975))
print(data.frame(wide[c("Dept", "Gender", "admitted")], low = limits[1, ], high = limits[2, ]))
```

An observed discrepancy: Department A

Table 23. Observed offers versus the 95% predictive interval under the common gender-effect model.

Department / gender	Applicants	Observed offers	95% predictive interval
A / male	825	512	[490, 564]
A / female	108	89	[60, 82]
E / female	393	94	[80, 124]

The observed 89 offers for women in A exceed the upper predictive limit (the observed proportion lies above its 95% predictive interval in the graphical check as well). One outlying cell among 12 could occur by chance, but it prompts a closer look: does a gender-by-department interaction improve the fit? Isolated deviations among 12 checks are possible.

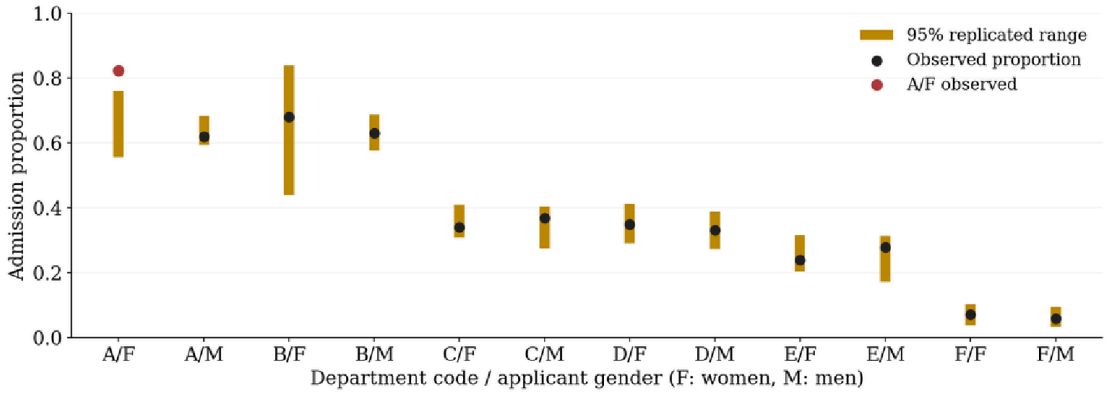


Figure 17. Posterior predictive check: observed admission proportion (dot) and 95% replicated range (bar) for each department/gender cell (F: women, M: men) under the common-effect model. The observed proportion for A/F (red) lies above its interval.

Posterior predictive distribution

For a new male applicant to department B, $Y_{\text{new},M,B}$, the probability of admission is

$$\Pr(Y_{\text{new},M,B} = 1 \mid y) = \int \frac{1}{1 + e^{-(\alpha+\gamma_B)}}\, \pi(\theta \mid y)\, d\theta \approx \frac{1}{S}\sum_{s=1}^{S} \frac{1}{1 + e^{-(\alpha^{(s)}+\gamma_B^{(s)})}}.$$

Similarly it is possible to derive the expected number of successful applications out of n_{new} applicants.

```
# Find the observed cell that defines the new applicant profile.
k <- which(wide$Dept == "A" & wide$Gender == "Female")
# Use the JAGS probabilities for that profile, one per posterior draw.
mat <- do.call(rbind, lapply(draws, as.matrix))
p_new <- mat[, paste0("p[", k, "]")]
# Conditional on each draw, simulate offers among 10 new applicants.
set.seed(20261028)
y_new <- rbinom(length(p_new), size = 10, prob = p_new)
# Report mean and central 95% range of the predictive distribution.
mean(y_new)
quantile(y_new, c(.025, .5, .975))
```

These predictions apply to a future group with the same recorded gender and department under an unchanged admissions process; a new cohort would be needed to assess out-of-sample accuracy. Illustrative output: mean ≈ 6.6; 95% predictive interval $[4, 9]$.

Table 24. The predictive answer depends on the fitted model: ten future women applying to Department A.

Model	Predictive mean	95% predictive interval
Department A women alone: Beta–Binomial	8.18	[5, 10]
All 12 cells: common-effect logistic model	≈ 6.6	≈ [4, 9]

The one-cell model estimates this group's rate from its 89 offers among 108 applicants. The joint model imposes one gender coefficient across departments; its replicated A/women counts tend to fall below the observed 89. Before using the joint model to forecast a new cohort, investigate that lack of fit and consider a department-specific gender effect.

Remark 9 (model fit versus predictive ability). **Model check:** simulate another 12-cell table with the same cell sizes, then compare counts, department rates and gender patterns with the observed table. **Predictive assessment:** use a later admissions cohort, held out before fitting, and evaluate calibration

or predictive log scores on those new data. A model can match the historical table yet predict a different future cohort poorly.

Gibbs sampling

The normal model with unknown mean and variance

The Gibbs sampler is a computational method to draw random samples of multidimensional parameters from their joint posterior. Again the posterior cannot be obtained in closed form; however, in some cases, for some parameters there exists a closed-form *conditional* posterior. Each parameter (or block of parameters) is then drawn sequentially from its conditional posterior.

Example: the distribution of income among blue-collar employees. Suppose the log-income Y is Normal with unknown mean μ and variance σ^2. Given a sample of $n = 40$ blue-collar employees: $Y_i \mid \mu, \sigma^2 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$; priors $\mu \sim N(m_0, s_0^2)$ and $\sigma^2 \sim \text{InverseGamma}(a_0, b_0)$. The joint posterior

$$\pi(\mu, \sigma^2 \mid y) \propto \underbrace{\pi(\mu)}_{N(m_0, s_0^2)} \; \underbrace{\pi(\sigma^2)}_{\text{IG}(a_0, b_0)} \; \underbrace{\prod_{i=1}^{n} f(y_i \mid \mu, \sigma^2)}_{\text{Likelihood}}$$

has no closed form. However, $\pi(\mu \mid \sigma^2, y)$ is a Normal density with variance and mean

$$V_\mu = \Big(\frac{1}{s_0^2} + \frac{n}{\sigma^2} \Big)^{-1}, \qquad M_\mu = V_\mu \Big(\frac{m_0}{s_0^2} + \frac{n \bar{y}}{\sigma^2} \Big),$$

and $\pi(\sigma^2 \mid \mu, y)$ is another inverse-gamma with shape $a_0 + n/2$ and scale $b_0 + \tfrac{1}{2}\sum_i (y_i - \mu)^2$. Thus μ and σ^2 are *conditionally conjugate* (for normally distributed data). The inverse-gamma with shape a and scale b has density proportional to $\pi(x) \propto x^{-a-1} \exp(-b/x)$ for $x > 0$.

Algorithm 10 (Gibbs sampler for (μ, σ^2)). Start at $(\mu^{(0)}, \sigma^{2(0)})$. At iteration s: draw $\mu^{(s)} \sim \pi(\mu \mid \sigma^{2(s-1)}, y)$, then $\sigma^{2(s)} \sim \pi(\sigma^2 \mid \mu^{(s)}, y)$. Use the old variance to draw a new mean; immediately use the new mean to draw a new variance. Repeat, retaining pairs after warm-up. JAGS can automate this process.

```
# Simulate 40 log annual incomes (euros) for the Gibbs example.
set.seed(41)
y <- rnorm(40, mean = 10, sd = 0.4)
n <- length(y)
# Set prior values and initialise the chain.
m0 <- 10; s0_sq <- 1; a0 <- 2; b0 <- 0.16
mu <- mean(y); sigma2 <- var(y)
draws <- matrix(NA_real_, nrow = 12000, ncol = 2)
# Alternate draws from the full conditional distributions.
for (s in 1:nrow(draws)) {
  V_mu <- 1 / (1 / s0_sq + n / sigma2)
  M_mu <- V_mu * (m0 / s0_sq + sum(y) / sigma2)
  mu <- rnorm(1, mean = M_mu, sd = sqrt(V_mu))
  shape <- a0 + n / 2
  rate <- b0 + sum((y - mu)^2) / 2
  sigma2 <- 1 / rgamma(1, shape = shape, rate = rate)
  draws[s, ] <- c(mu, sigma2)
}
```

The same model in JAGS

```
# The same data y, n and priors as in the manual Gibbs example.
normal_model <- "model {
  for (i in 1:N) { y[i] ~ dnorm(mu, precision) }
  mu ~ dnorm(m0, 1 / s0_sq)
  precision ~ dgamma(a0, b0)
  sigma2 <- 1 / precision
  mean_income <- exp(mu + sigma2 / 2)
}"
tc <- textConnection(normal_model)
```

```
jm <- rjags::jags.model(tc, data = list(
  N = length(y), y = y, m0 = 10, s0_sq = 1,
  a0 = 2, b0 = 0.16), n.chains = 4, n.adapt = 1000)
close(tc)
update(jm, 1000)
draws <- rjags::coda.samples(jm, c("mu", "sigma2", "mean_income"), n.iter = 4000)
summary(draws)                          # Compare with the hand-coded Gibbs draws.
```

A gamma prior with shape a_0 and rate b_0 for the precision $1/\sigma^2$ corresponds to the stated inverse-gamma prior for σ^2 (shape a_0, scale b_0). The trace plots of the Gibbs run show the mean log-income moving around 10 (first 1,000 of 10,000 retained iterations of an independent illustrative run); after warm-up, check stable exploration: a single chain cannot establish convergence, so compare several chains in practice.

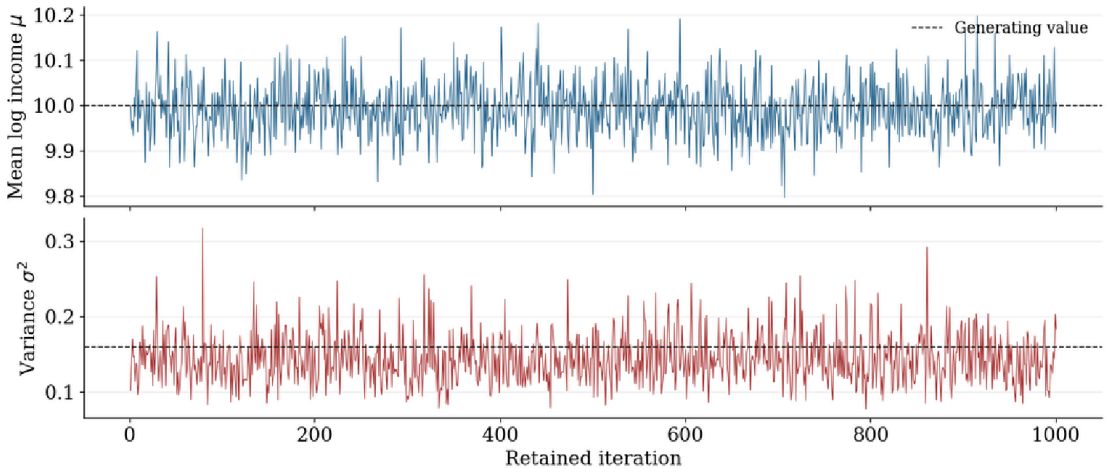


Figure 18. Gibbs traces for the log-income example: mean log-income μ (top) and variance σ^2 (bottom), first 1,000 of 10,000 retained iterations of an independent illustrative run; the dashed lines mark the generating values.

For each retained joint draw $(\mu^{(s)}, \sigma^{2(s)})$ the conditional mean annual income is $e^{\mu^{(s)}+\sigma^{2(s)}/2}$. Summarise those transformed draws rather than exponentiating a marginal interval for μ.

Table 25. Posterior summaries: log-income and annual income (rounded values from a separate simulation; μ and σ^2 refer to log-euros).

Parameter	Posterior median	Illustrative 95% interval
Mean log-income μ	9.99	[9.87, 10.11]
Variance σ^2	0.14	[0.09, 0.22]
Mean income $e^{\mu+\sigma^2/2}$	EUR 23,400	[EUR 20,900, EUR 26,800]

Hamiltonian Monte Carlo, NUTS and Stan

More complex models

Suppose we had individual Berkeley application records: applicant i's admission $Y_i \in \{0,1\}$, gender F_i, department d_i, prior grade x_i and school $s(i)$. An illustrative extension is

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = \alpha + \beta_F F_i + \gamma_{d_i} + \delta x_i + u_{s(i)}, \quad u_s = \tau z_s,\ z_s \sim N(0,1),$$

with $\gamma_A = 0$. x_i records a measured applicant feature; the latent u_s captures remaining differences among schools; $\tau > 0$ controls their spread. With many schools the posterior has many related z_s as

well as common coefficients. Gradient-based HMC can explore such a posterior effectively; the non-centred form can help its geometry.

Hamiltonian Monte Carlo: use the posterior gradient

Sequential samples from a posterior obtained with Metropolis–Hastings and Gibbs sampling (when possible) are likely to be correlated when the parameter space is high-dimensional. Hamiltonian Monte Carlo (partially) overcomes this by using a proposal based on the gradient of the log-posterior, which allows a more efficient exploration of the parameter space and better mixing across chains with different starting points.

In the hypothetical applicant-level model let θ collect all unknown continuous parameters (including z_s and τ) and y the observed decisions. HMC draws an artificial momentum $r \sim N(0, M)$ of the same dimension as θ and defines the total energy

$$H(\theta, r) = U(\theta) + K(r), \qquad U(\theta) = -\log \pi(\theta \mid y), \qquad K(r) = \tfrac{1}{2} r^\top M^{-1} r,$$

where H is the total energy, U the negative log posterior density, K the kinetic energy and M a positive-definite scale matrix; again the posterior normalising constant cancels. The posterior gradient gives a direction; numerical steps move θ and r together to propose a distant value of θ. An accept/reject step corrects the numerical approximation error. This can move more efficiently than an unguided random walk, particularly with many correlated parameters. HMC needs a differentiable posterior density in continuous parameters (see the Stan Reference Manual).

NUTS and Stan

Stan, with its R interface RStan, draws samples from a complex posterior. It implements the No-U-Turn Sampler (NUTS), an adaptive form of HMC requiring little or no user input; it translates a probability model into a log density and gradients and runs NUTS by default. Users still choose the model, priors, parameterisation and data checks. Diagnostics to inspect: $\hat{R}$, ESS, trace plots.

Stan from R: the practical workflow

Stan is the modelling language and inference engine: specify data, unknown continuous parameters, priors and likelihood in a `.stan` program. RStan and CmdStanR are two R interfaces; the lecture uses CmdStanR, which compiles the program through a separate CmdStan installation and returns draws to R. Build a named R list whose entries match the Stan `data` block; compile with `cmdstan_model()`; sample four NUTS chains with `$sample()`; inspect summaries and sampler warnings; extract coefficient draws to calculate posterior odds ratios, intervals and predictions, as with JAGS.

```
# Run once: install the R interface from the Stan repository.
install.packages("cmdstanr", repos = c(
  "https://stan-dev.r-universe.dev", getOption("repos")))
# A working C++ toolchain is needed to compile Stan models.
cmdstanr::check_cmdstan_toolchain()
# Install the separate CmdStan engine, then verify it is available.
cmdstanr::install_cmdstan(cores = 2)
cmdstanr::cmdstan_version()
```

```
data {
  int<lower=1> K;
  array[K] int<lower=0> admitted;
  array[K] int<lower=0> applicants;
  array[K] int<lower=0, upper=1> female;
  array[K] int<lower=1, upper=6> dept;
}
parameters {
  real alpha;
  real beta_female;
  vector[5] beta_dept;
}
transformed parameters {
  vector[K] eta;
  for (k in 1:K) {
```

```
    eta[k] = alpha + beta_female * female[k];
    if (dept[k] > 1) eta[k] += beta_dept[dept[k] - 1];
  }
}
model {
  alpha ~ normal(0, 2.5);
  beta_female ~ normal(0, 2.5);
  beta_dept ~ normal(0, 2.5);
  admitted ~ binomial_logit(applicants, eta);
}
generated quantities { real OR_female = exp(beta_female); }
```

The integer arrays describe the 12 grouped cells; Department A is the reference and the five-element vector stores the effects of Departments B–F. The likelihood is Binomial on the count scale; Stan uses the logit parameterisation for numerical stability, and the generated quantity converts the gender coefficient to an odds ratio for every posterior draw.

```
# The complete Stan program is stan_code in the companion R script.
stan_file <- tempfile(fileext = ".stan")
writeLines(stan_code, stan_file)                  # Write the Stan language model.
stan_model <- cmdstanr::cmdstan_model(stan_file)   # Compile it.
# Fit four NUTS chains with warm-up, then inspect diagnostics.
fit <- stan_model$sample(data = stan_data, seed = 20261026,
                         chains = 4, parallel_chains = 4,
                         iter_warmup = 1000, iter_sampling = 2000)
fit$summary(variables = c("beta_female", "OR_female"))   # R-hat, ESS, intervals.
fit$diagnostic_summary()                                  # Divergences and other sampler warnings.

# Flatten retained chains to a matrix of posterior draws.
mat <- posterior::as_draws_matrix(fit$draws(
  variables = c("alpha", "beta_female", "OR_female")))
# Report an equal-tail interval and the probability of a negative contrast.
quantile(mat[, "OR_female"], c(.025, .5, .975))
mean(mat[, "beta_female"] < 0)
# Probability for a new female applicant to Department A.
p_A <- plogis(mat[, "alpha"] + mat[, "beta_female"])
# For each posterior state, draw a count among ten similar applicants.
y_new <- rbinom(length(p_A), size = 10, prob = p_A)
quantile(y_new, c(.025, .5, .975))
```

The R package `cmdstanr`, CmdStan and a working C++ toolchain are required. Each prediction propagates the coefficient uncertainty and the outcome variation, as in the earlier posterior predictive formula.

Divergences are a model diagnostic. A divergent transition means the numerical trajectory may not have followed the target distribution accurately in some region. Inspect where they occur; consider stronger prior information, rescaling, or a better parameterisation. Refit and recheck $\hat{R}$, ESS, trace plots and Monte Carlo error. For a low-dimensional example like this, a well-fitting model should usually sample without divergences; the diagnostics must still be checked.

Choosing a computational approach and practical workflow

Table 26. Choosing a computational approach.

Method	Main idea	Best teaching point
Conjugate update	Algebraic posterior in the same family	Exact Beta–Binomial analysis; no sampler needed
Metropolis–Hastings	Propose a move and accept or reject it	General and transparent; proposal tuning matters
Gibbs	Draw each parameter from its full conditional	Conditional conjugacy can help; mixing may be slow
HMC / NUTS	Gradient-guided proposals	Efficient for continuous models; diagnose divergences

A practical Bayesian workflow

1. **Question and model:** define the probability model for the data and the statistical research question.
2. **Before fitting:** simulate the quantities of interest from random parameter draws from their prior distribution and revise implausible assumptions.
3. **Computation:** fit the posterior; check trace plots, $\hat{R}$, effective sample size and Monte Carlo error.
4. **Interpretation** of the results.
5. **Revisit and report:** refit, check the sensitivity to the prior, especially with smaller samples, and assess prediction on a held-out part of the data.

Recap and takeaways

- Bayes' rule combines a likelihood and a prior to ground the analysis on the posterior distribution.
- The posterior supports estimation, uncertainty statements and prediction.
- Conjugacy can give an exact posterior; MCMC approximates it when direct calculation is difficult.
- Metropolis–Hastings, Gibbs and HMC use different transition strategies; JAGS and Stan turn a model specification into posterior draws. The ultimate judgement of the outcome is still the analyst's.
- For the Berkeley table, adjustment for department changes the question from a pooled comparison to a model-based within-department association.

Further topics: large datasets and variational inference

MCMC replaces a difficult integral by averages of posterior draws, but obtaining those draws can be costly. Each iteration may evaluate a likelihood (and, for HMC, its gradient) over a large dataset; the chain progresses sequentially; warm-up, slow mixing and multiple chains add computation. With many observations and latent variables the time needed for sufficiently accurate posterior summaries may be substantial. This motivates approximate inference by optimisation, among other approaches; the useful comparison is the computational cost against the accuracy needed for the scientific question.

Variational inference: approximate by optimisation

Choose a tractable family $q_\phi(\theta)$ and find the parameter value that makes $q_\phi(\theta)$ as close as possible to the posterior by minimising

$$\mathrm{KL}\big(q_\phi(\theta) \,\|\, \pi(\theta \mid y)\big) = \mathbb{E}_{q_\phi}\Big[\log \frac{q_\phi(\theta)}{\pi(\theta \mid y)}\Big].$$

Because the posterior normalising constant does not depend on ϕ, this is equivalent to maximising the evidence lower bound (ELBO),

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(y, \theta)] - \mathbb{E}_{q_\phi}[\log q_\phi(\theta)].$$

Speed can improve, but restrictive families may miss dependence across elements of θ or understate uncertainty; check the approximation when it matters (Blei, Kucukelbir and McAuliffe, 2017).

Example: topic models. Suppose we have a large collection of documents. In latent Dirichlet allocation (LDA) each document mixes topics and each topic assigns probabilities to words; the topic proportions and assignments are latent. The inferential task is to estimate which topics occur in each document and which words characterise each topic; the computational challenge is that jointly updating all these latent quantities across many documents is difficult. Blei, Ng and Jordan (2003) used

variational methods for approximate inference in LDA; at larger scale, stochastic variational inference uses subsets of documents per update, and Hoffman et al. (2013) analysed corpora with millions of documents.

References (Part IV)

- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A. and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd ed. CRC Press.
- Gelman, A., Vehtari, A., McElreath, R. et al. (2026). *Bayesian Workflow*. Chapman & Hall/CRC Press.
- Steorts, R. C. (2017). STA 360/601: Bayesian Methods and Modern Statistics. Lecture slides, Modules 1 and 5–7. Duke University.
- Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H. and Teller, E. (1953). Equation of state calculations by fast computing machines. *Journal of Chemical Physics*, 21, 1087–1092.
- Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57, 97–109.
- Geman, S. and Geman, D. (1984). Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE TPAMI*, 6, 721–741.
- Bickel, P. J., Hammel, E. A. and O'Connell, J. W. (1975). Sex bias in graduate admissions: data from Berkeley. *Science*, 187, 398–404.
- Plummer, M. (2003). JAGS: a program for analysis of Bayesian graphical models using Gibbs sampling. *Proceedings of DSC 2003*.
- Carpenter, B. et al. (2017). Stan: a probabilistic programming language. *Journal of Statistical Software*, 76(1).
- Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15, 1593–1623.
- McElreath, R. (2020). *Statistical Rethinking*, 2nd ed. CRC Press.
- Blei, D. M., Ng, A. Y. and Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Hoffman, M. D., Blei, D. M., Wang, C. and Paisley, J. (2013). Stochastic variational inference. *Journal of Machine Learning Research*, 14, 1303–1347.
- Blei, D. M., Kucukelbir, A. and McAuliffe, J. D. (2017). Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112, 859–877.
- Documentation: R `datasets::UCBAdmissions` (search.r-project.org); JAGS (mcmc-jags.sourceforge.io); Stan diagnostics (mc-stan.org/learn-stan/diagnostics-warnings.html) and CmdStanR (mc-stan.org/cmdstanr).